GRC102 Week 4 Practical Lab

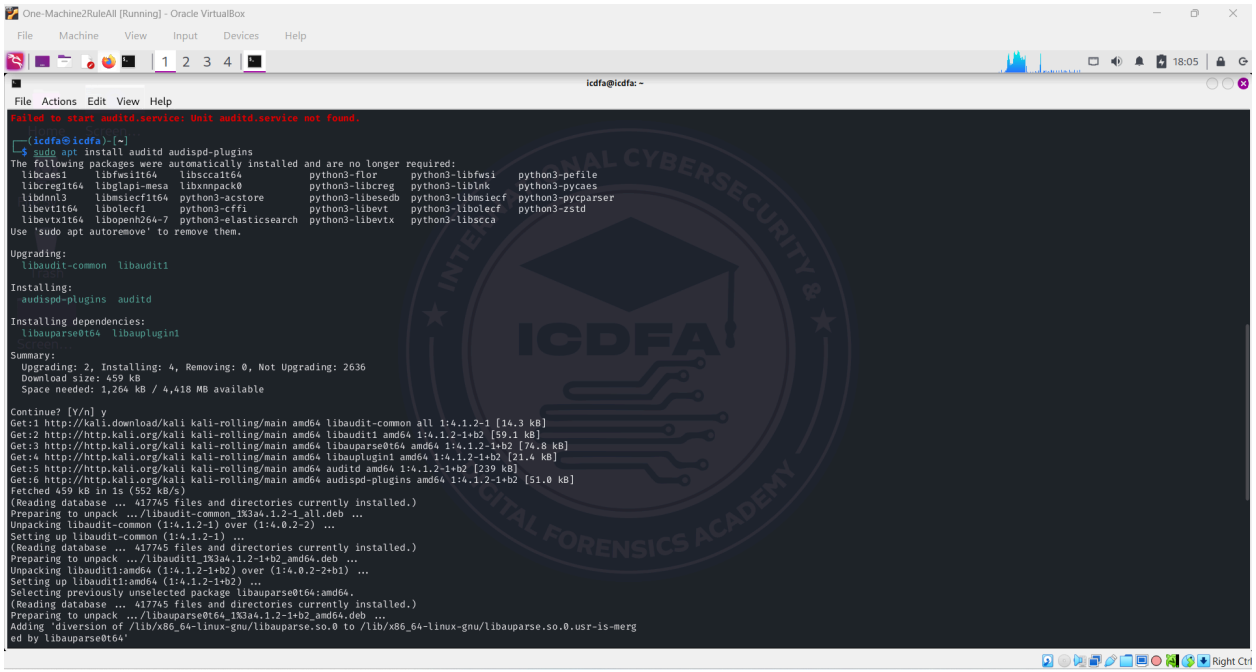
Linux Security Monitoring and Auditing

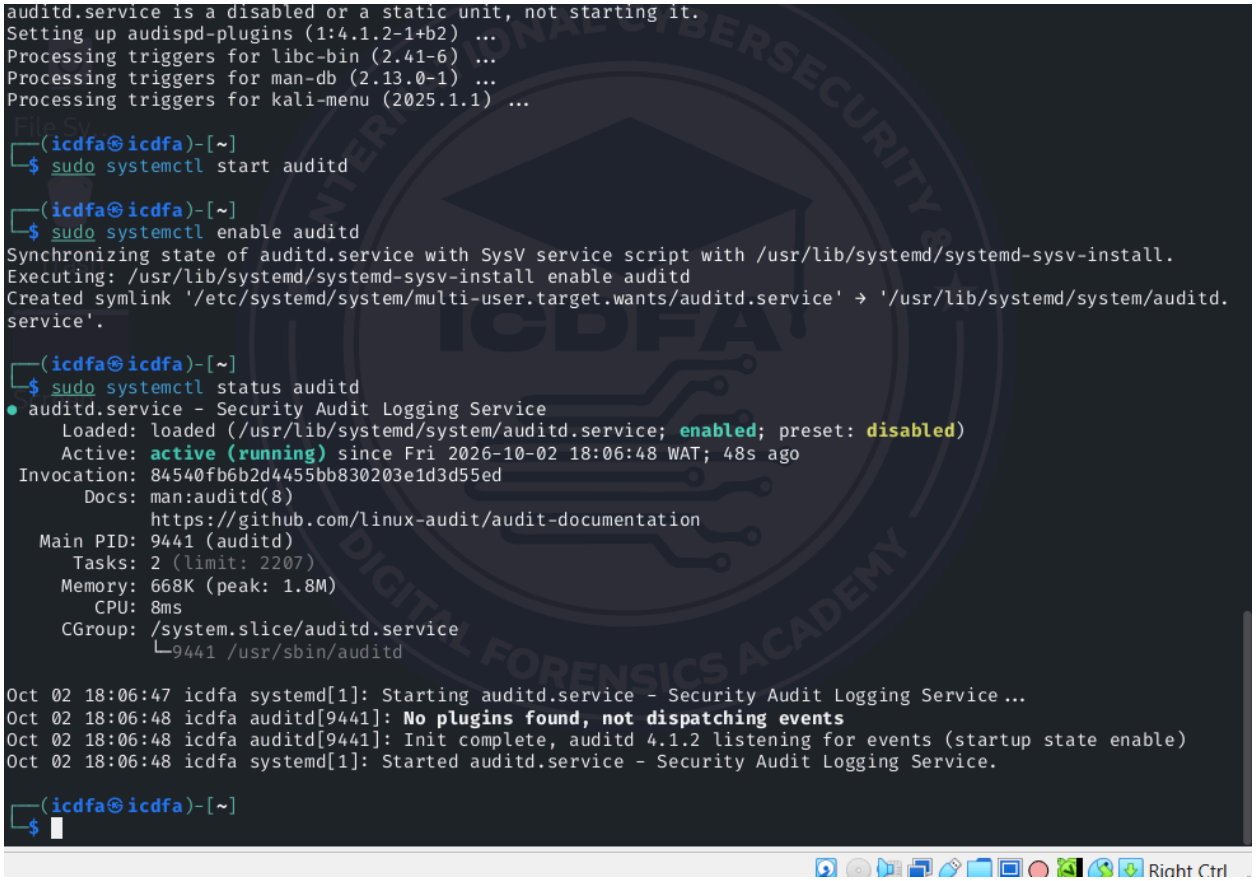
From Technical Evidence to Governance Assurance

Field	Detail
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Registration Details	C11/26/CGRCE/17603 ,Cohort 11 CGRCE
Date	2nd October, 2026

Activity 1




One-Machine2RuleAll [Running] - Oracle VirtualBox
File Machine View Input Devices Help
icdfa@icdfa: ~
File Actions Edit View Help
Failed to start auditd.service: Unit auditd.service not found.
┌──(icdfa㉿icdfa)-[~]
└─$ sudo apt install auditd audispd-plugins
The following packages were automatically installed and are no longer required:
libcaes1 libfwsi1t64 libsccal1t64 python3-flor python3-libfwsi python3-pefile
libcreg1t64 libglapi-mesa libxnnpack0 python3-libcreg python3-liblnk python3-pycaes
libdnnl3 libmsiecf1t64 python3-acstore python3-libesedb python3-libmsiecf python3-pycparser
libevt1t64 libolecf1 python3-cffi python3-libevt python3-libolecf python3-zstd
libevtx1t64 libopenh264-7 python3-elasticsearch python3-libevtx python3-libscca
Use 'sudo apt autoremove' to remove them.
Upgrading:
libaudit-common libaudit1
Installing:
audispd-plugins auditd
Installing dependencies:
libauparse0t64 libauplugin1
Summary:
Upgrading: 2, Installing: 4, Removing: 0, Not Upgrading: 2636
Download size: 459 kB
Space needed: 1,264 kB / 4,418 MB available
Continue? [Y/n] y
Get:1 http://kali.download/kali kali-rolling/main amd64 libaudit-common all 1:4.1.2-1 [14.3 kB]
Get:2 http://http.kali.org/kali kali-rolling/main amd64 libaudit1 amd64 1:4.1.2-1+b2 [59.1 kB]
Get:3 http://http.kali.org/kali kali-rolling/main amd64 libauparse0t64 amd64 1:4.1.2-1+b2 [74.8 kB]
Get:4 http://http.kali.org/kali kali-rolling/main amd64 libauplugin1 amd64 1:4.1.2-1+b2 [21.4 kB]
Get:5 http://http.kali.org/kali kali-rolling/main amd64 auditd amd64 1:4.1.2-1+b2 [239 kB]
Get:6 http://http.kali.org/kali kali-rolling/main amd64 audispd-plugins amd64 1:4.1.2-1+b2 [51.0 kB]
Fetched 459 kB in 1s (552 kB/s)
(Reading database ... 417745 files and directories currently installed.)
Preparing to unpack .../libaudit-common_1%3a4.1.2-1_all.deb ...
Unpacking libaudit-common (1:4.1.2-1) over (1:4.0.2-2) ...
Setting up libaudit-common (1:4.1.2-1) ...
(Reading database ... 417745 files and directories currently installed.)
Preparing to unpack .../libaudit1_1%3a4.1.2-1+b2_amd64.deb ...
Unpacking libaudit1:amd64 (1:4.1.2-1+b2) over (1:4.0.2-2+b1) ...
Setting up libaudit1:amd64 (1:4.1.2-1+b2) ...
Selecting previously unselected package libauparse0t64:amd64.
(Reading database ... 417745 files and directories currently installed.)
Preparing to unpack .../libauparse0t64_1%3a4.1.2-1+b2_amd64.deb ...
Adding 'diversion of /lib/x86_64-linux-gnu/libauparse.so.0 to /lib/x86_64-linux-gnu/libauparse.so.0.usr-is-merged by libauparse0t64'
Right Ctrl


auditd.service is a disabled or a static unit, not starting it.
Setting up audispd-plugins (1:4.1.2-1+b2) ...
Processing triggers for libc-bin (2.41-6) ...
Processing triggers for man-db (2.13.0-1) ...
Processing triggers for kali-menu (2025.1.1) ...
┌──(icdfa㉿icdfa)-[~]
└─$ sudo systemctl start auditd
┌──(icdfa㉿icdfa)-[~]
└─$ sudo systemctl enable auditd
Synchronizing state of auditd.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable auditd
Created symlink '/etc/systemd/system/multi-user.target.wants/auditd.service' → '/usr/lib/systemd/system/auditd.service'.
┌──(icdfa㉿icdfa)-[~]
└─$ sudo systemctl status auditd
● auditd.service - Security Audit Logging Service
Loaded: loaded (/usr/lib/systemd/system/auditd.service; enabled; preset: disabled)
Active: active (running) since Fri 2026-10-02 18:06:48 WAT; 48s ago
Invocation: 84540fb6b2d4455bb830203e1d3d55ed
Docs: man:auditd(8)
https://github.com/linux-audit/audit-documentation
Main PID: 9441 (auditd)
Tasks: 2 (limit: 2207)
Memory: 668K (peak: 1.8M)
CPU: 8ms
CGroup: /system.slice/auditd.service
└─9441 /usr/sbin/auditd
Oct 02 18:06:47 icdfa systemd[1]: Starting auditd.service - Security Audit Logging Service...
Oct 02 18:06:48 icdfa auditd[9441]: No plugins found, not dispatching events
Oct 02 18:06:48 icdfa auditd[9441]: Init complete, auditd 4.1.2 listening for events (startup state enable)
Oct 02 18:06:48 icdfa systemd[1]: Started auditd.service - Security Audit Logging Service.
┌──(icdfa㉿icdfa)-[~]
└─$
Right Ctrl

```
└─9441 /usr/sbin/auditd

Oct 02 18:06:47 icdfa systemd[1]: Starting auditd.service - Security Audit Logging Service...
Oct 02 18:06:48 icdfa auditd[9441]: No plugins found, not dispatching events
Oct 02 18:06:48 icdfa auditd[9441]: Init complete, auditd 4.1.2 listening for events (startup state enable)
Oct 02 18:06:48 icdfa systemd[1]: Started auditd.service - Security Audit Logging Service.

┌──(icdfa㉿icdfa)-[~]
└─$ sudo nano /etc/audit/rules.d/custom.rules

┌──(icdfa㉿icdfa)-[~]
└─$ -w /etc/passwd -p rwxa -k passwd_changes
-w: command not found

┌──(icdfa㉿icdfa)-[~]
└─$ sudo nano /etc/audit/rules.d/custom.rules

┌──(icdfa㉿icdfa)-[~]
└─$ sudo nano /etc/audit/rules.d/custom.rules

┌──(icdfa㉿icdfa)-[~]
└─$ sudo systemctl restart auditd

┌──(icdfa㉿icdfa)-[~]
└─$ sudo auditctl -l
-w /etc/passwd -p rwxa -k passwd_changes

┌──(icdfa㉿icdfa)-[~]
└─$
```

Right Ctrl

icdfa@icdfa: ~

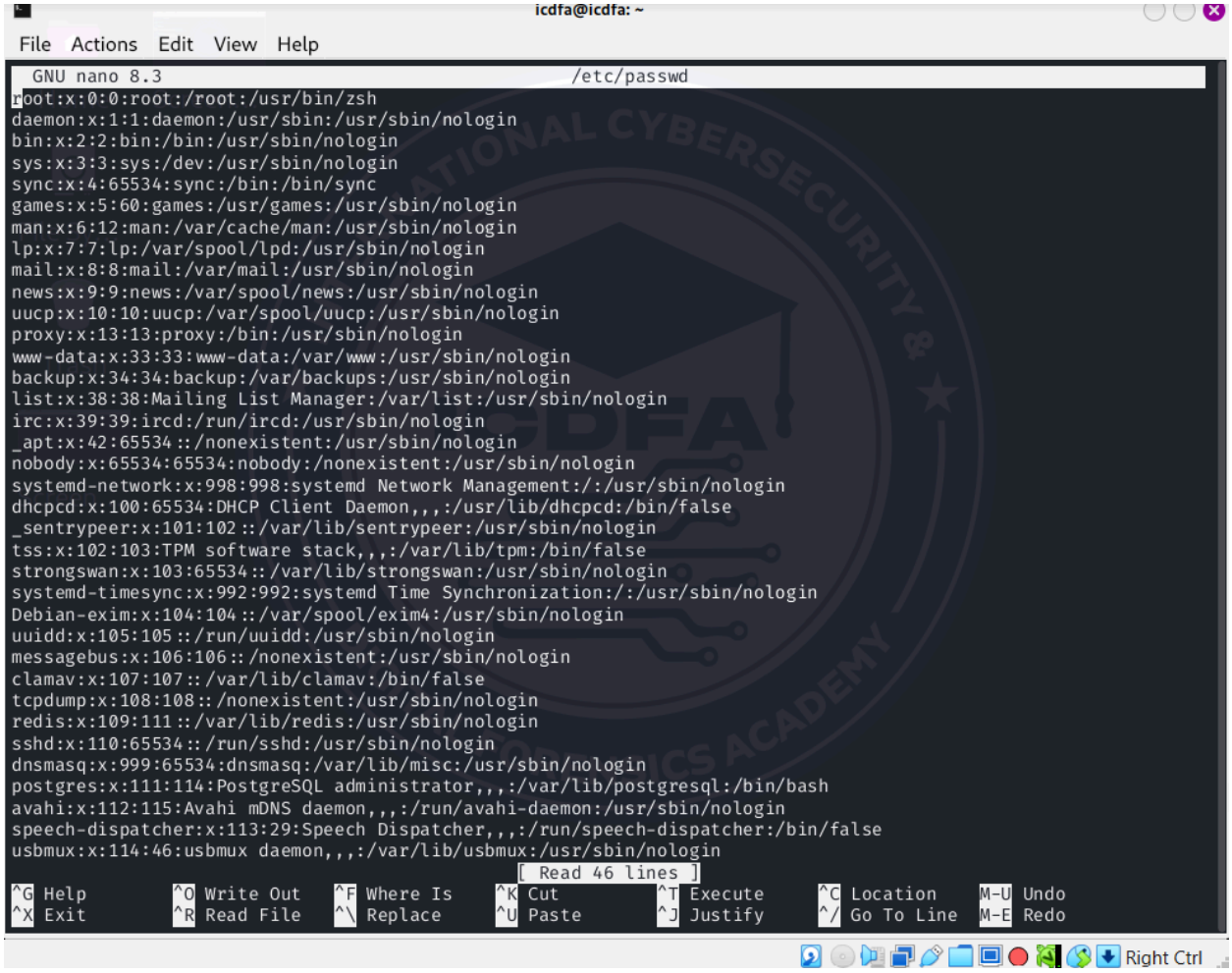
File Actions Edit View Help

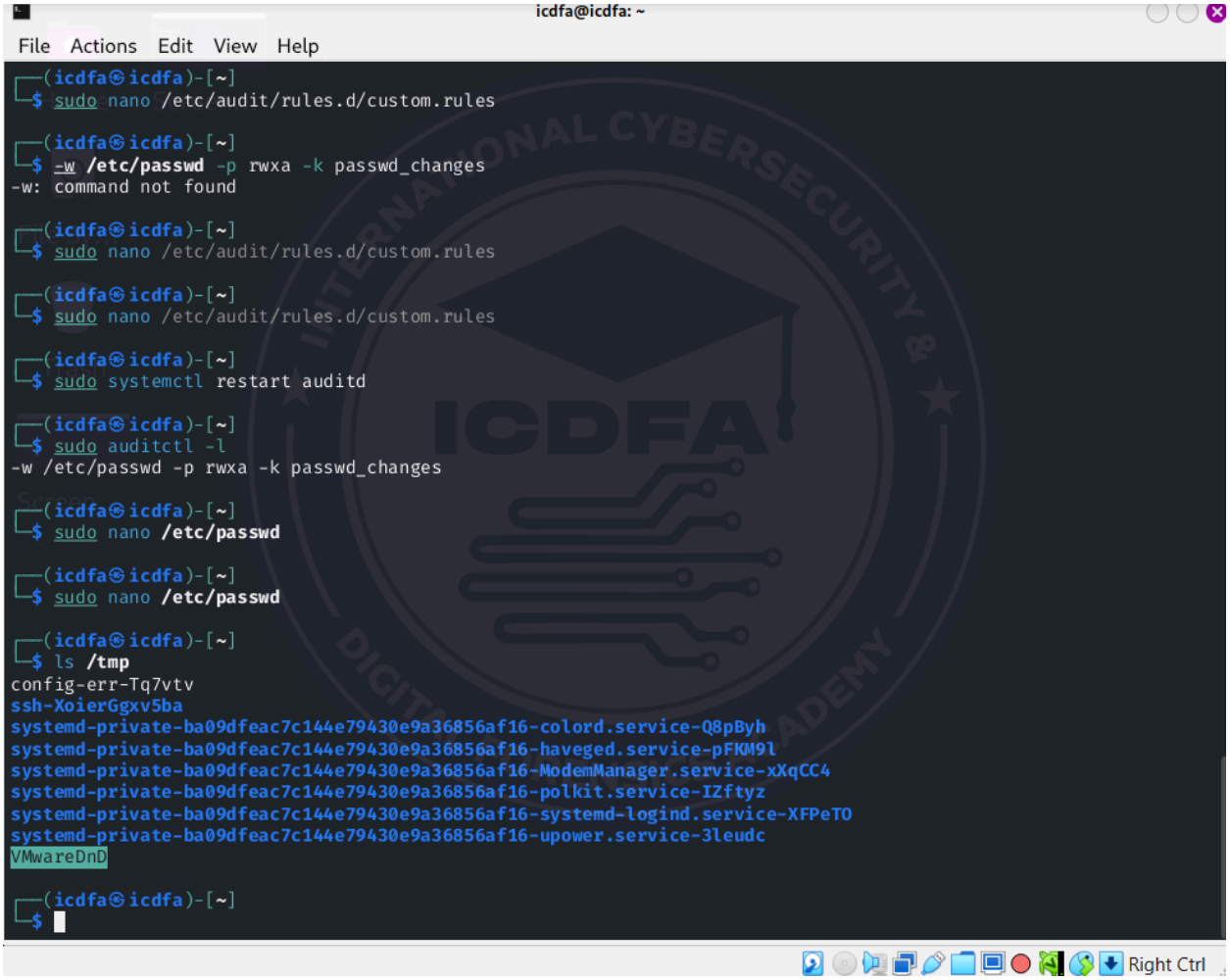
```
 GNU nano 8.3                                    /etc/passwd
root:x:0:0:root:/root:/usr/bin/zsh
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:/:/usr/sbin/nologin
dhcpcd:x:100:65534:DHCP Client Daemon,,,:/usr/lib/dhcpcd:/bin/false
_sentrypeer:x:101:102::/var/lib/sentrypeer:/usr/sbin/nologin
tss:x:102:103:TPM software stack,,,:/var/lib/tpm:/bin/false
strongswan:x:103:65534::/var/lib/strongswan:/usr/sbin/nologin
systemd-timesync:x:992:992:systemd Time Synchronization:/:/usr/sbin/nologin
Debian-exim:x:104:104::/var/spool/exim4:/usr/sbin/nologin
uuidd:x:105:105::/run/uuidd:/usr/sbin/nologin
messagebus:x:106:106::/nonexistent:/usr/sbin/nologin
clamav:x:107:107::/var/lib/clamav:/bin/false
tcpdump:x:108:108::/nonexistent:/usr/sbin/nologin
redis:x:109:111::/var/lib/redis:/usr/sbin/nologin
sshd:x:110:65534::/run/sshd:/usr/sbin/nologin
dnsmasq:x:999:65534:dnsmasq:/var/lib/misc:/usr/sbin/nologin
postgres:x:111:114:PostgreSQL administrator,,,:/var/lib/postgresql:/bin/bash
avahi:x:112:115:Avahi mDNS daemon,,,:/run/avahi-daemon:/usr/sbin/nologin
speech-dispatcher:x:113:29:Speech Dispatcher,,,:/run/speech-dispatcher:/bin/false
usbmux:x:114:46:usbmux daemon,,,:/var/lib/usbmux:/usr/sbin/nologin
                                [ Read 46 lines ]
^G Help      ^O Write Out   ^F Where Is    ^K Cut         ^T Execute     ^C Location    M-U Undo
^X Exit      ^R Read File   ^\ Replace     ^U Paste       ^J Justify     ^/ Go To Line  M-E Redo
```

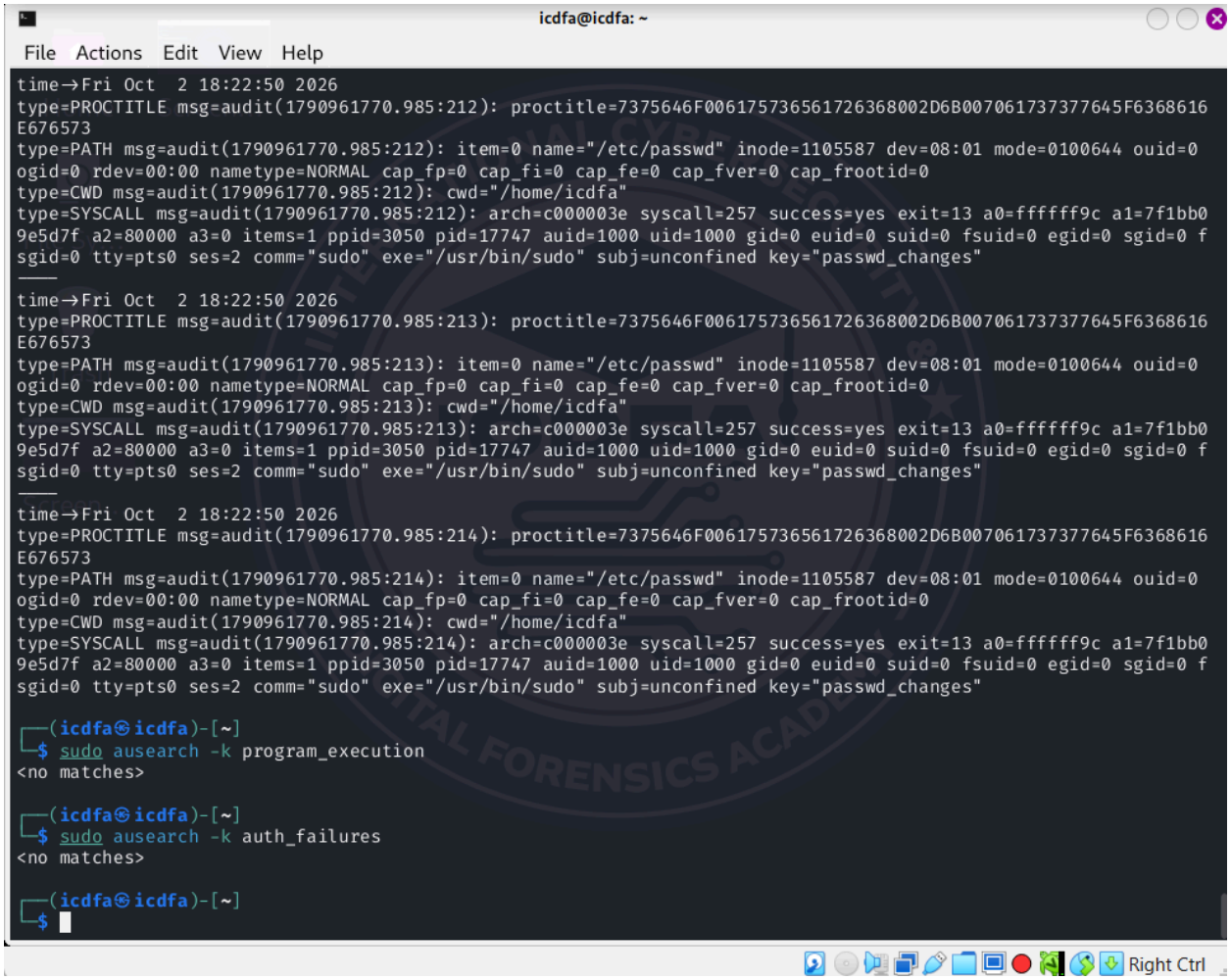
Right Ctrl


ed by libauparse0t64'
Adding 'diversion of /lib/x86_64-linux-gnu/libauparse.so.0.0.0 to /lib/x86_64-linux-gnu/libauparse.so.0.0.0.usr
-is-merged by libauparse0t64'
Unpacking libauparse0t64:amd64 (1:4.1.2-1+b2) ...
Selecting previously unselected package libauplugin1:amd64.
Preparing to unpack .../libauplugin1_1%3a4.1.2-1+b2_amd64.deb ...
Unpacking libauplugin1:amd64 (1:4.1.2-1+b2) ...
Selecting previously unselected package auditd.
Preparing to unpack .../auditd_1%3a4.1.2-1+b2_amd64.deb ...
Unpacking auditd (1:4.1.2-1+b2) ...
Selecting previously unselected package audispd-plugins.
Preparing to unpack .../audispd-plugins_1%3a4.1.2-1+b2_amd64.deb ...
Unpacking audispd-plugins (1:4.1.2-1+b2) ...
Setting up libauparse0t64:amd64 (1:4.1.2-1+b2) ...
Setting up libauplugin1:amd64 (1:4.1.2-1+b2) ...
Setting up auditd (1:4.1.2-1+b2) ...
update-rc.d: We have no instructions for the auditd init script.
update-rc.d: It looks like a non-network service, we enable it.
audit-rules.service is a disabled or a static unit, not starting it.
auditd.service is a disabled or a static unit, not starting it.
Setting up audispd-plugins (1:4.1.2-1+b2) ...
Processing triggers for libc-bin (2.41-6) ...
Processing triggers for man-db (2.13.0-1) ...
Processing triggers for kali-menu (2025.1.1) ...
┌──(icdfa㉿icdfa)-[~]
└─$
Right Ctrl


1 2 3 4
18:16
icdfa@icdfa: ~
File Actions Edit View Help
GNU nano 8.3 /etc/audit/rules.d/custom.rules
-w /etc/passwd -p rwxa -k passwd_changes
-w /etc/shadow -p rwxa shadow_changes
-a always,exit -F arcch=b64 -S execve -k program_execution
-a always,exit -F arch=b32 -S execve -k program_execution
-w /var/log/auth.log -p wa -k auth_failures
Trash
Screen...
[Read 8 lines]
^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute ^C Location M-U Undo
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line M-E Redo
Right Ctrl


icdfa@icdfa: ~
File Actions Edit View Help
GNU nano 8.3 /etc/passwd
root:x:0:0:root:/root:/usr/bin/zsh
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:/:/usr/sbin/nologin
dhcpcd:x:100:65534:DHCP Client Daemon,,,:/usr/lib/dhcpcd:/bin/false
_sentrypeer:x:101:102::/var/lib/sentrypeer:/usr/sbin/nologin
tss:x:102:103:TPM software stack,,,:/var/lib/tpm:/bin/false
strongswan:x:103:65534::/var/lib/strongswan:/usr/sbin/nologin
systemd-timesync:x:992:992:systemd Time Synchronization:/:/usr/sbin/nologin
Debian-exim:x:104:104::/var/spool/exim4:/usr/sbin/nologin
uuidd:x:105:105::/run/uuidd:/usr/sbin/nologin
messagebus:x:106:106::/nonexistent:/usr/sbin/nologin
clamav:x:107:107::/var/lib/clamav:/bin/false
tcpdump:x:108:108::/nonexistent:/usr/sbin/nologin
redis:x:109:111::/var/lib/redis:/usr/sbin/nologin
sshd:x:110:65534::/run/sshd:/usr/sbin/nologin
dnsmasq:x:999:65534:dnsmasq:/var/lib/misc:/usr/sbin/nologin
postgres:x:111:114:PostgreSQL administrator,,,:/var/lib/postgresql:/bin/bash
avahi:x:112:115:Avahi mDNS daemon,,,:/run/avahi-daemon:/usr/sbin/nologin
speech-dispatcher:x:113:29:Speech Dispatcher,,,:/run/speech-dispatcher:/bin/false
usbmux:x:114:46:usbmux daemon,,,:/var/lib/usbmux:/usr/sbin/nologin
[Read 46 lines]
^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute ^C Location M-U Undo
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line M-E Redo
Right Ctrl


icdfa@icdfa: ~
File Actions Edit View Help
┌──(icdfa㉿icdfa)-[~]
└─$ sudo nano /etc/audit/rules.d/custom.rules
┌──(icdfa㉿icdfa)-[~]
└─$ -w /etc/passwd -p rwxa -k passwd_changes
-w: command not found
┌──(icdfa㉿icdfa)-[~]
└─$ sudo nano /etc/audit/rules.d/custom.rules
┌──(icdfa㉿icdfa)-[~]
└─$ sudo nano /etc/audit/rules.d/custom.rules
┌──(icdfa㉿icdfa)-[~]
└─$ sudo systemctl restart auditd
┌──(icdfa㉿icdfa)-[~]
└─$ sudo auditctl -l
-w /etc/passwd -p rwxa -k passwd_changes
┌──(icdfa㉿icdfa)-[~]
└─$ sudo nano /etc/passwd
┌──(icdfa㉿icdfa)-[~]
└─$ sudo nano /etc/passwd
┌──(icdfa㉿icdfa)-[~]
└─$ ls /tmp
config-err-Tq7vtv
ssh-XoierGgxv5ba
systemd-private-ba09dfeac7c144e79430e9a36856af16-colord.service-Q8pByh
systemd-private-ba09dfeac7c144e79430e9a36856af16-haveged.service-pFKM9l
systemd-private-ba09dfeac7c144e79430e9a36856af16-ModemManager.service-xXqCC4
systemd-private-ba09dfeac7c144e79430e9a36856af16-polkit.service-IZftyz
systemd-private-ba09dfeac7c144e79430e9a36856af16-systemd-logind.service-XFPeTO
systemd-private-ba09dfeac7c144e79430e9a36856af16-upower.service-3leudc
VMwareDnD
┌──(icdfa㉿icdfa)-[~]
└─$
Right Ctrl


icdfa@icdfa: ~
File Actions Edit View Help
time→Fri Oct 2 18:22:50 2026
type=PROCTITLE msg=audit(1790961770.985:212): proctitle=7375646F006175736561726368002D6B007061737377645F6368616E676573
type=PATH msg=audit(1790961770.985:212): item=0 name="/etc/passwd" inode=1105587 dev=08:01 mode=0100644 ouid=0 ogid=0 rdev=00:00 nametype=NORMAL cap_fp=0 cap_fi=0 cap_fe=0 cap_fver=0 cap_frootid=0
type=CWD msg=audit(1790961770.985:212): cwd="/home/icdfa"
type=SYSCALL msg=audit(1790961770.985:212): arch=c000003e syscall=257 success=yes exit=13 a0=ffffff9c a1=7f1bb09e5d7f a2=80000 a3=0 items=1 ppid=3050 pid=17747 auid=1000 uid=1000 gid=0 euid=0 suid=0 fsuid=0 egid=0 sgid=0 fsgid=0 tty=pts0 ses=2 comm="sudo" exe="/usr/bin/sudo" subj=unconfined key="passwd_changes"
time→Fri Oct 2 18:22:50 2026
type=PROCTITLE msg=audit(1790961770.985:213): proctitle=7375646F006175736561726368002D6B007061737377645F6368616E676573
type=PATH msg=audit(1790961770.985:213): item=0 name="/etc/passwd" inode=1105587 dev=08:01 mode=0100644 ouid=0 ogid=0 rdev=00:00 nametype=NORMAL cap_fp=0 cap_fi=0 cap_fe=0 cap_fver=0 cap_frootid=0
type=CWD msg=audit(1790961770.985:213): cwd="/home/icdfa"
type=SYSCALL msg=audit(1790961770.985:213): arch=c000003e syscall=257 success=yes exit=13 a0=ffffff9c a1=7f1bb09e5d7f a2=80000 a3=0 items=1 ppid=3050 pid=17747 auid=1000 uid=1000 gid=0 euid=0 suid=0 fsuid=0 egid=0 sgid=0 fsgid=0 tty=pts0 ses=2 comm="sudo" exe="/usr/bin/sudo" subj=unconfined key="passwd_changes"
time→Fri Oct 2 18:22:50 2026
type=PROCTITLE msg=audit(1790961770.985:214): proctitle=7375646F006175736561726368002D6B007061737377645F6368616E676573
type=PATH msg=audit(1790961770.985:214): item=0 name="/etc/passwd" inode=1105587 dev=08:01 mode=0100644 ouid=0 ogid=0 rdev=00:00 nametype=NORMAL cap_fp=0 cap_fi=0 cap_fe=0 cap_fver=0 cap_frootid=0
type=CWD msg=audit(1790961770.985:214): cwd="/home/icdfa"
type=SYSCALL msg=audit(1790961770.985:214): arch=c000003e syscall=257 success=yes exit=13 a0=ffffff9c a1=7f1bb09e5d7f a2=80000 a3=0 items=1 ppid=3050 pid=17747 auid=1000 uid=1000 gid=0 euid=0 suid=0 fsuid=0 egid=0 sgid=0 fsgid=0 tty=pts0 ses=2 comm="sudo" exe="/usr/bin/sudo" subj=unconfined key="passwd_changes"
(icdfa㉿icdfa)-[~]
$ sudo ausearch -k program_execution
<no matches>
(icdfa㉿icdfa)-[~]
$ sudo ausearch -k auth_failures
<no matches>
(icdfa㉿icdfa)-[~]
$
Right Ctrl


icdfa@icdfa: ~
File Actions Edit View Help
(icdfa㉿icdfa)-[~]
$ sudo aureport
Summary Report
Range of time in logs: 02/10/2026 18:06:48.002 - 02/10/2026 18:24:19.669
Selected time for report: 02/10/2026 18:06:48 - 02/10/2026 18:24:19.669
Number of changes in configuration: 7
Number of changes to accounts, groups, or roles: 0
Number of logins: 0
Number of failed logins: 0
Number of authentications: 0
Number of failed authentications: 0
Number of users: 3
Number of terminals: 7
Number of host names: 2
Number of executables: 8
Number of commands: 8
Number of files: 1
Number of AVC's: 0
Number of MAC events: 0
Number of failed syscalls: 4
Number of anomaly events: 0
Number of responses to anomaly events: 0
Number of crypto events: 0
Number of integrity events: 0
Number of virt events: 0
Number of keys: 1
Number of process IDs: 35
Number of events: 267
(icdfa㉿icdfa)-[~]
$ sudo aureport --failed
Failed Summary Report
Range of time in logs: 02/10/2026 18:06:48.002 - 02/10/2026 18:24:33.896
Selected time for report: 02/10/2026 18:06:48 - 02/10/2026 18:24:33.896
Number of changes in configuration: 0
Number of changes to accounts, groups, or roles: 0
Number of logins: 0
Number of failed logins: 0
Number of authentications: 0
Number of failed authentications: 0
Number of users: 2
Number of terminals: 2
Right Ctrl


icdfa@icdfa: ~
File Actions Edit View Help
(icdfa㉿icdfa)-[~]
$ sudo aureport --failed
Failed Summary Report
Range of time in logs: 02/10/2026 18:06:48.002 - 02/10/2026 18:24:33.896
Selected time for report: 02/10/2026 18:06:48 - 02/10/2026 18:24:33.896
Number of changes in configuration: 0
Number of changes to accounts, groups, or roles: 0
Number of logins: 0
Number of failed logins: 0
Number of authentications: 0
Number of failed authentications: 0
Number of users: 2
Number of terminals: 2
Number of host names: 1
Number of executables: 2
Number of commands: 2
Number of files: 1
Number of AVC's: 0
Number of MAC events: 0
Number of failed syscalls: 4
Number of anomaly events: 0
Number of responses to anomaly events: 0
Number of crypto events: 0
Number of integrity events: 0
Number of virt events: 0
Number of keys: 1
Number of process IDs: 3
Number of events: 5
(icdfa㉿icdfa)-[~]
$ sudo aureport --login
Login Report
date time auid host term exe success event
<no events of interest were found>
(icdfa㉿icdfa)-[~]
$

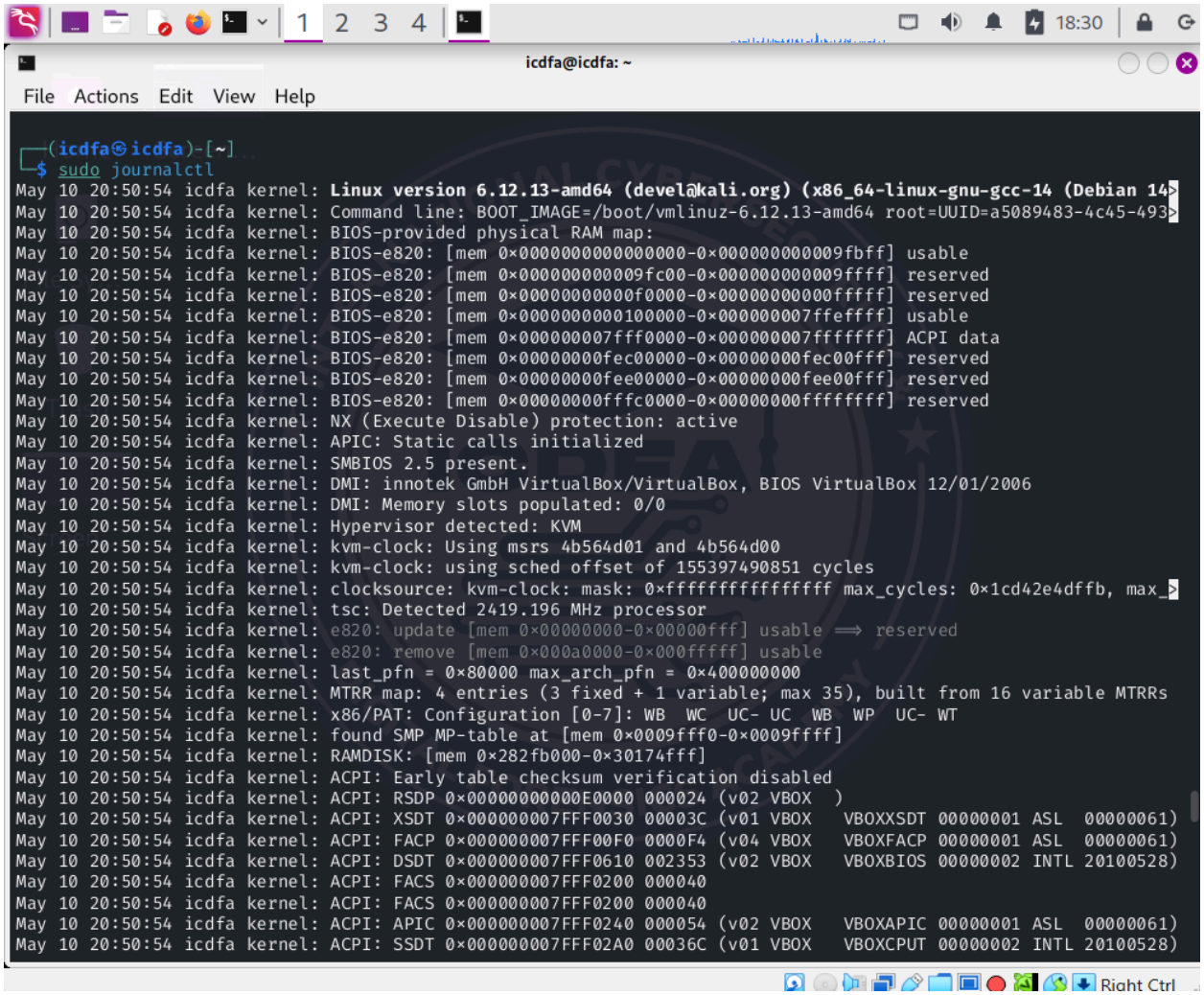
Report

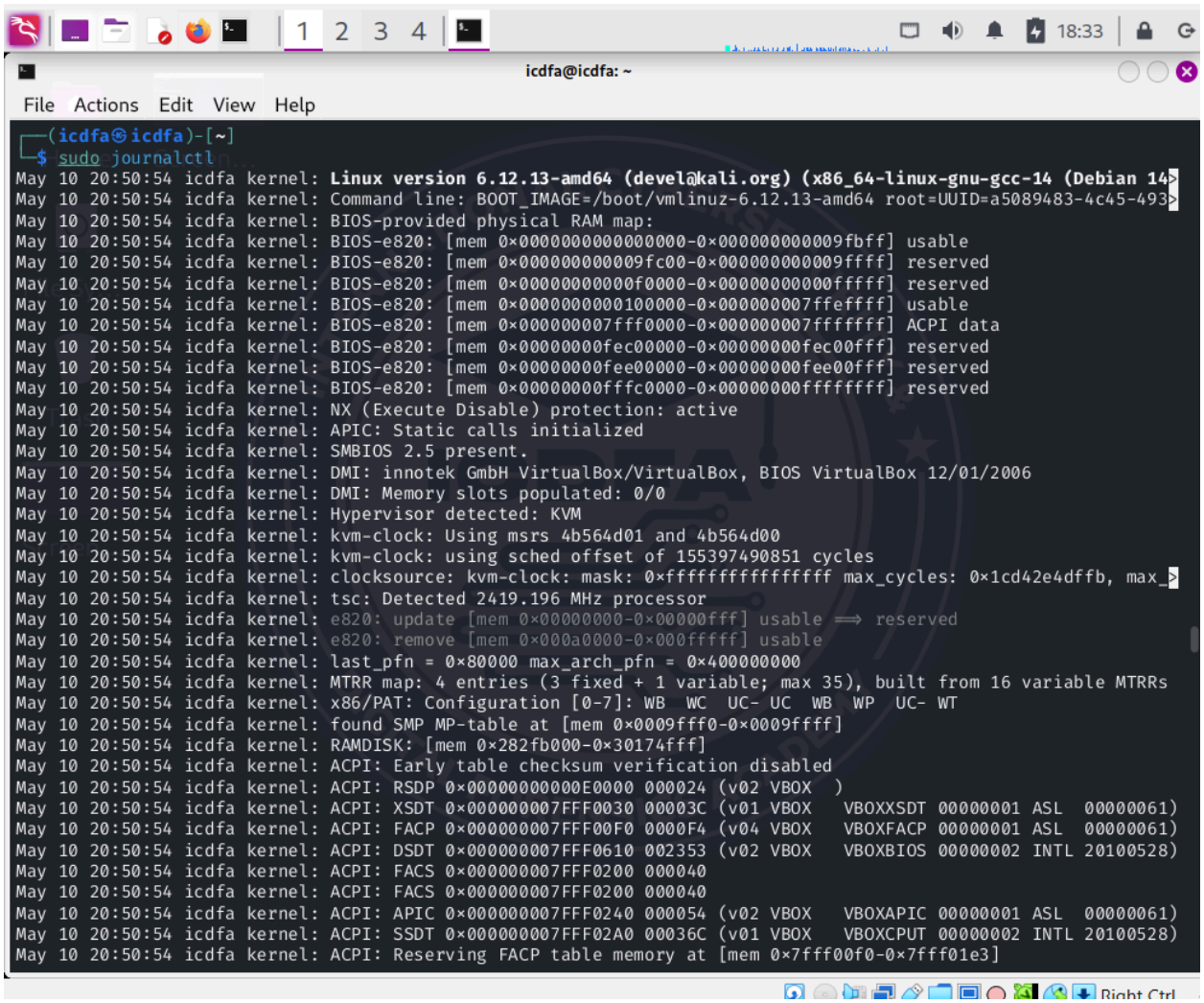
Log Analysis (Primary Event Details)

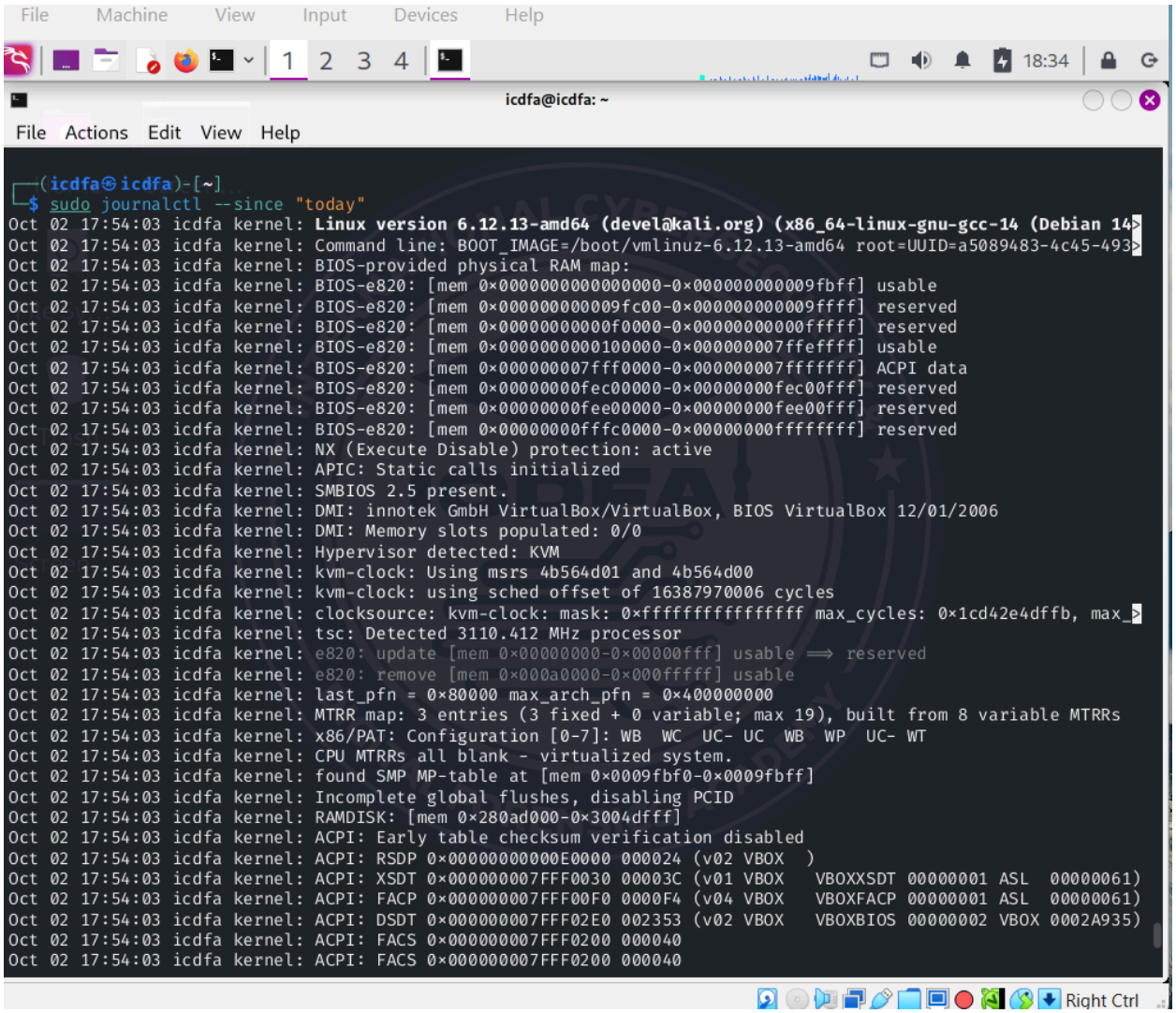
- **Identity & Context:** `auid=1000` alongside `uid=1000` identifies activity by user icdfa. The attributes `euid=0` and `egid=0` confirm execution with root authority via `comm="sudo"` (binary `exe="/usr/bin/sudo"`). Preserving the original `auid` guarantees precise attribution across privilege transitions.
- **Target Operation:** A successful `openat` system call (`syscall=257`, `success=yes`, `exit=13`) targeting `/etc/passwd` (`name="/etc/passwd"`, `mode=0100644`, root ownership). The configured watch triggered under `key="passwd_changes"`. Decoding the hex string in `proctitle` reveals `sudo nano /etc/passwd`, demonstrating editor access aligned with Image 6 where `sudo nano /etc/passwd` was executed twice.
- **Timing & Session:** Captured Fri Oct 2 2026 at 18:22:50 on pseudo-terminal `pts0` (session `ses=2`, path `/home/icdfa`). Multiple records within the same millisecond (IDs 212–214) represent distinct log items generated by one open call.

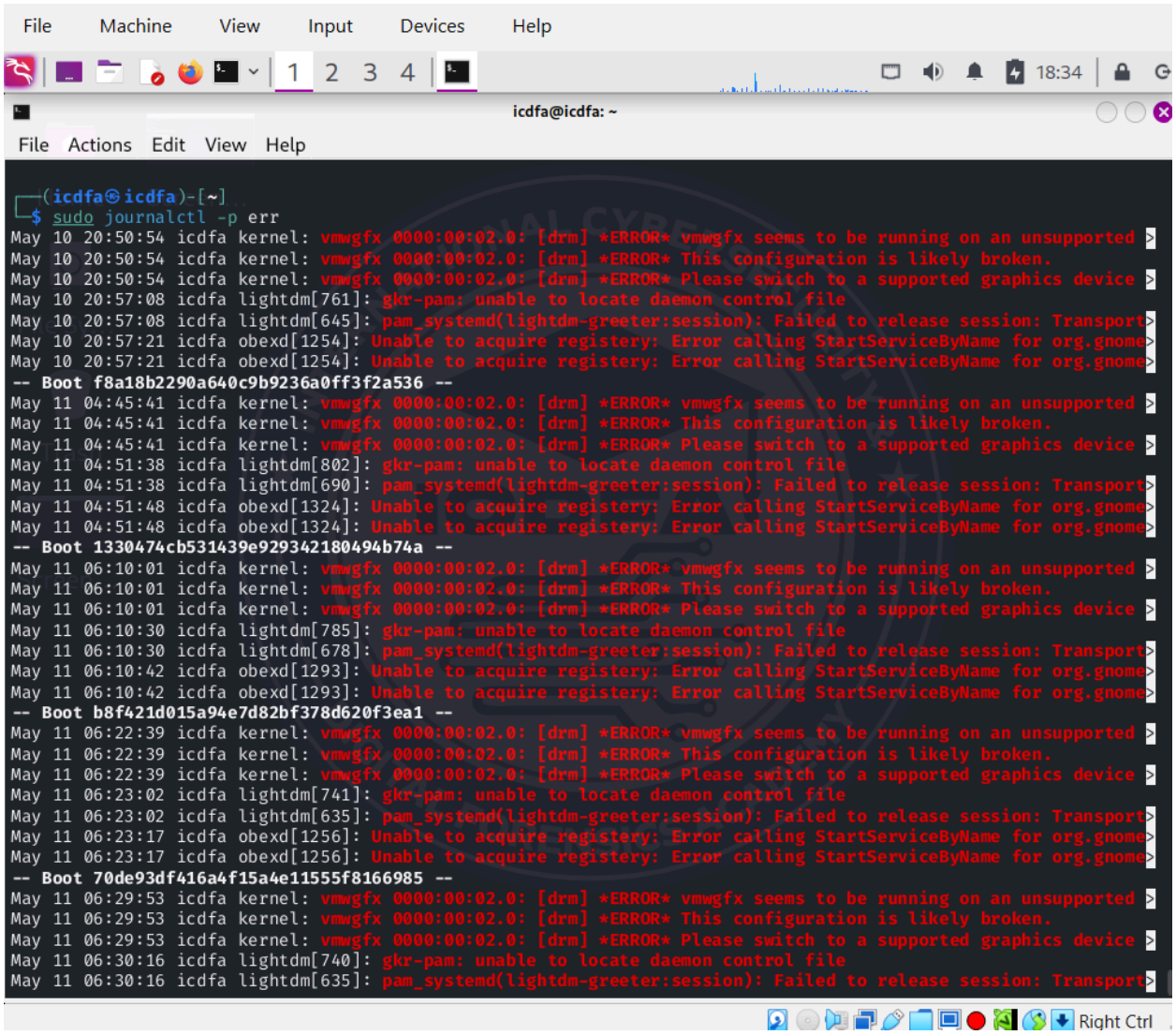
Governance & Risk Implications

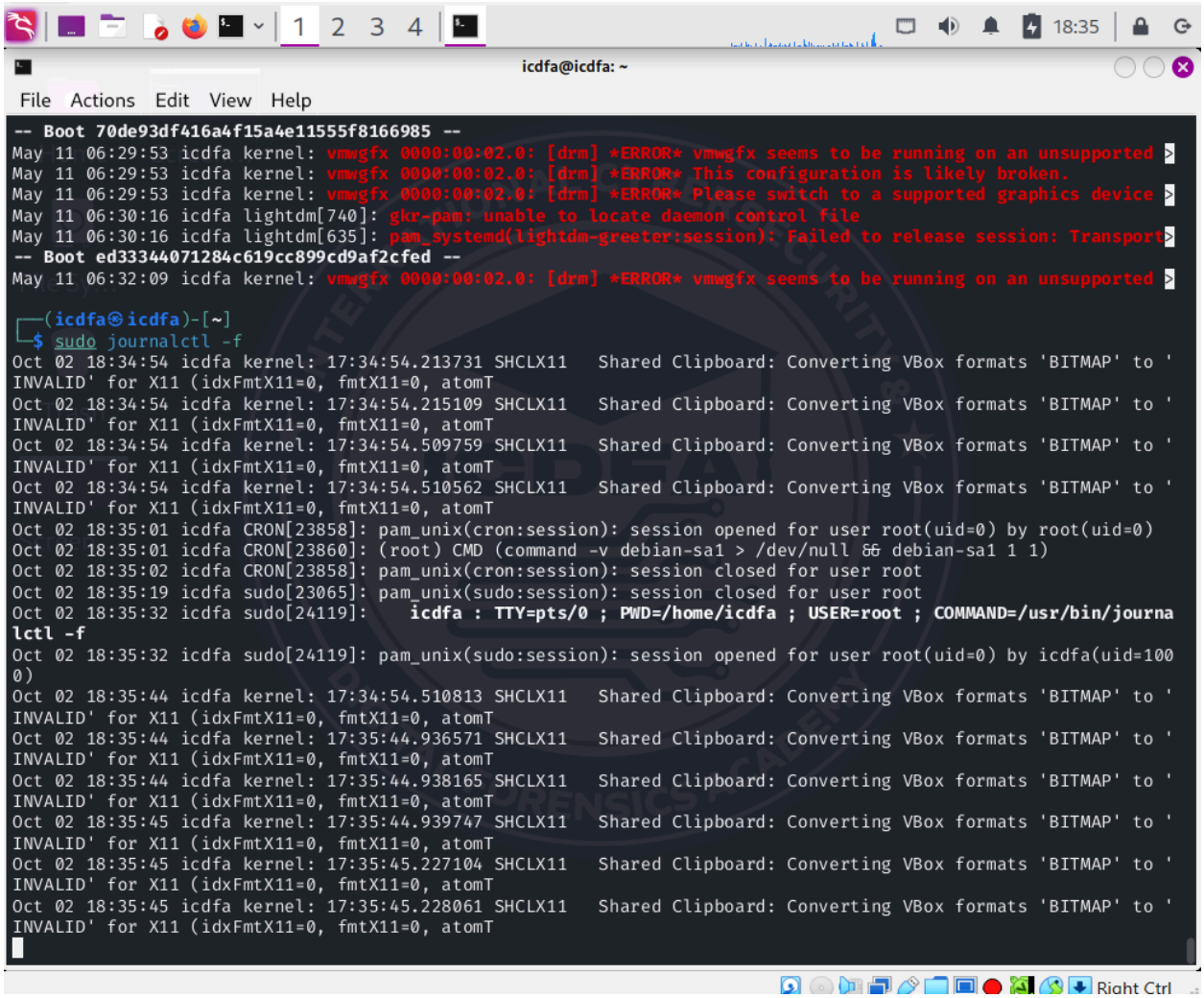
- Because `/etc/passwd` manages accounts, UIDs, and default shells, unapproved modifications (like creating root-equivalent accounts or altering shells) represent severe persistence risks.
- Audit logging ensures non-repudiation by tying timestamped actions and specific keys to designated identities. This baseline telemetry validates authorization protocols, internal audit requirements, and access controls.
- Because the `rwxa` rule captures read attempts, this telemetry confirms file access rather than confirmed modification. Analysts should perform hash comparisons or check `aureport` for account-level modifications.
- Summary data in Image 8 indicates 267 total events, 7 configuration changes, 0 authentication failures, 0 account modifications, and 4 failed syscalls. The zero account modifications metric confirms the file was inspected rather than saved with changes.
- Queries for `program_execution` and `auth_failures` yielded `<no matches>` in Image 7 due to two key issues:
    - Syntax error in rule configuration (`arcch=b64` instead of `arch=b64`) prevented proper rule loading, while the `-F arch=b32` statement was rejected on pure 64-bit architecture.
    - The file `/var/log/auth.log` is absent on Kali instances using systemd-journald, rendering the file watch inactive.
- Unmonitored gaps present substantial governance risk by offering false assurance; rule loading must be validated using `auditctl -l` and functional tests.

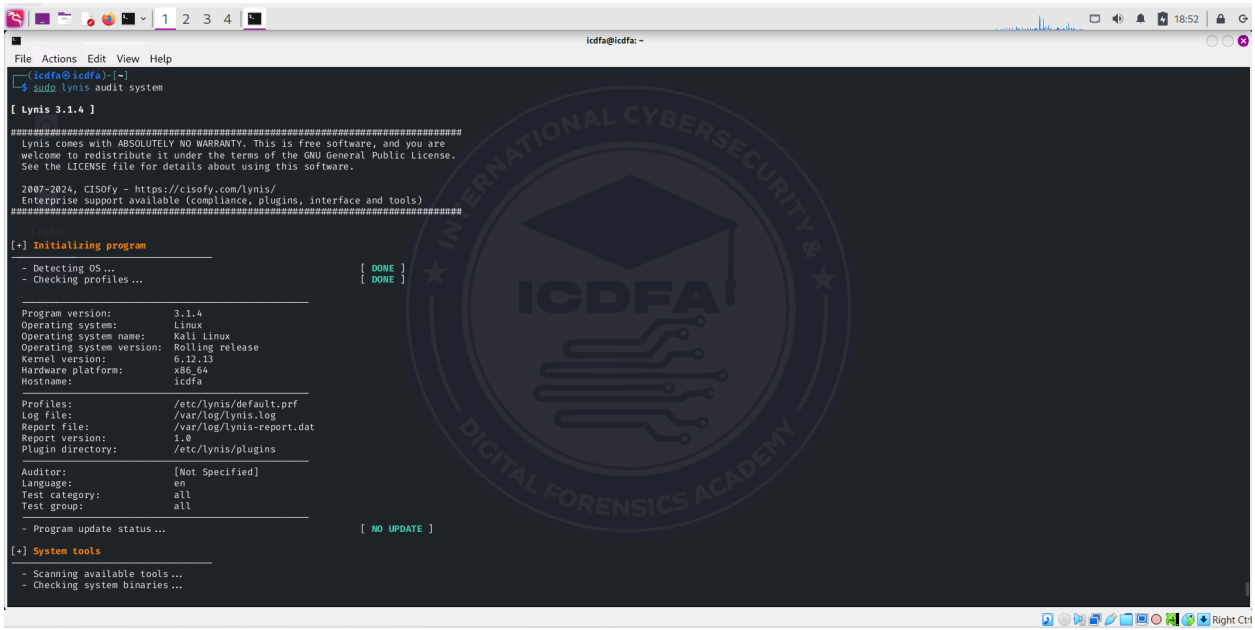

1 2 3 4
18:30
icdfa@icdfa: ~
File Actions Edit View Help
(icdfa㉿icdfa)-[~]
$ sudo journalctl
May 10 20:50:54 icdfa kernel: Linux version 6.12.13-amd64 (devel@kali.org) (x86_64-linux-gnu-gcc-14 (Debian 14>
May 10 20:50:54 icdfa kernel: Command line: BOOT_IMAGE=/boot/vmlinuz-6.12.13-amd64 root=UUID=a5089483-4c45-493>
May 10 20:50:54 icdfa kernel: BIOS-provided physical RAM map:
May 10 20:50:54 icdfa kernel: BIOS-e820: [mem 0x0000000000000000-0x000000000009fbff] usable
May 10 20:50:54 icdfa kernel: BIOS-e820: [mem 0x000000000009fc00-0x000000000009ffff] reserved
May 10 20:50:54 icdfa kernel: BIOS-e820: [mem 0x00000000000f0000-0x00000000000fffff] reserved
May 10 20:50:54 icdfa kernel: BIOS-e820: [mem 0x0000000000100000-0x000000007ffeffff] usable
May 10 20:50:54 icdfa kernel: BIOS-e820: [mem 0x000000007fff0000-0x000000007fffffff] ACPI data
May 10 20:50:54 icdfa kernel: BIOS-e820: [mem 0x00000000fec00000-0x00000000fec00fff] reserved
May 10 20:50:54 icdfa kernel: BIOS-e820: [mem 0x00000000fee00000-0x00000000fee00fff] reserved
May 10 20:50:54 icdfa kernel: BIOS-e820: [mem 0x00000000fffc0000-0x00000000ffffffff] reserved
May 10 20:50:54 icdfa kernel: NX (Execute Disable) protection: active
May 10 20:50:54 icdfa kernel: APIC: Static calls initialized
May 10 20:50:54 icdfa kernel: SMBIOS 2.5 present.
May 10 20:50:54 icdfa kernel: DMI: innotek GmbH VirtualBox/VirtualBox, BIOS VirtualBox 12/01/2006
May 10 20:50:54 icdfa kernel: DMI: Memory slots populated: 0/0
May 10 20:50:54 icdfa kernel: Hypervisor detected: KVM
May 10 20:50:54 icdfa kernel: kvm-clock: Using msrs 4b564d01 and 4b564d00
May 10 20:50:54 icdfa kernel: kvm-clock: using sched offset of 155397490851 cycles
May 10 20:50:54 icdfa kernel: clocksource: kvm-clock: mask: 0xffffffffffffffff max_cycles: 0x1cd42e4dffb, max_>
May 10 20:50:54 icdfa kernel: tsc: Detected 2419.196 MHz processor
May 10 20:50:54 icdfa kernel: e820: update [mem 0x00000000-0x00000fff] usable ==> reserved
May 10 20:50:54 icdfa kernel: e820: remove [mem 0x000a0000-0x000fffff] usable
May 10 20:50:54 icdfa kernel: last_pfn = 0x80000 max_arch_pfn = 0x400000000
May 10 20:50:54 icdfa kernel: MTRR map: 4 entries (3 fixed + 1 variable; max 35), built from 16 variable MTRRs
May 10 20:50:54 icdfa kernel: x86/PAT: Configuration [0-7]: WB WC UC- UC WB WP UC- WT
May 10 20:50:54 icdfa kernel: found SMP MP-table at [mem 0x0009fff0-0x0009ffff]
May 10 20:50:54 icdfa kernel: RAMDISK: [mem 0x282fb000-0x30174fff]
May 10 20:50:54 icdfa kernel: ACPI: Early table checksum verification disabled
May 10 20:50:54 icdfa kernel: ACPI: RSDP 0x00000000000E0000 000024 (v02 VBOX)
May 10 20:50:54 icdfa kernel: ACPI: XSDT 0x000000007FFF0030 00003C (v01 VBOX VBOXXSDT 00000001 ASL 00000061)
May 10 20:50:54 icdfa kernel: ACPI: FACP 0x000000007FFF00F0 0000F4 (v04 VBOX VBOXFACP 00000001 ASL 00000061)
May 10 20:50:54 icdfa kernel: ACPI: DSDT 0x000000007FFF0610 002353 (v02 VBOX VBOXBIOS 00000002 INTL 20100528)
May 10 20:50:54 icdfa kernel: ACPI: FACS 0x000000007FFF0200 000040
May 10 20:50:54 icdfa kernel: ACPI: FACS 0x000000007FFF0200 000040
May 10 20:50:54 icdfa kernel: ACPI: APIC 0x000000007FFF0240 000054 (v02 VBOX VBOXAPIC 00000001 ASL 00000061)
May 10 20:50:54 icdfa kernel: ACPI: SSDT 0x000000007FFF02A0 00036C (v01 VBOX VBOXCPUT 00000002 INTL 20100528)
Right Ctrl

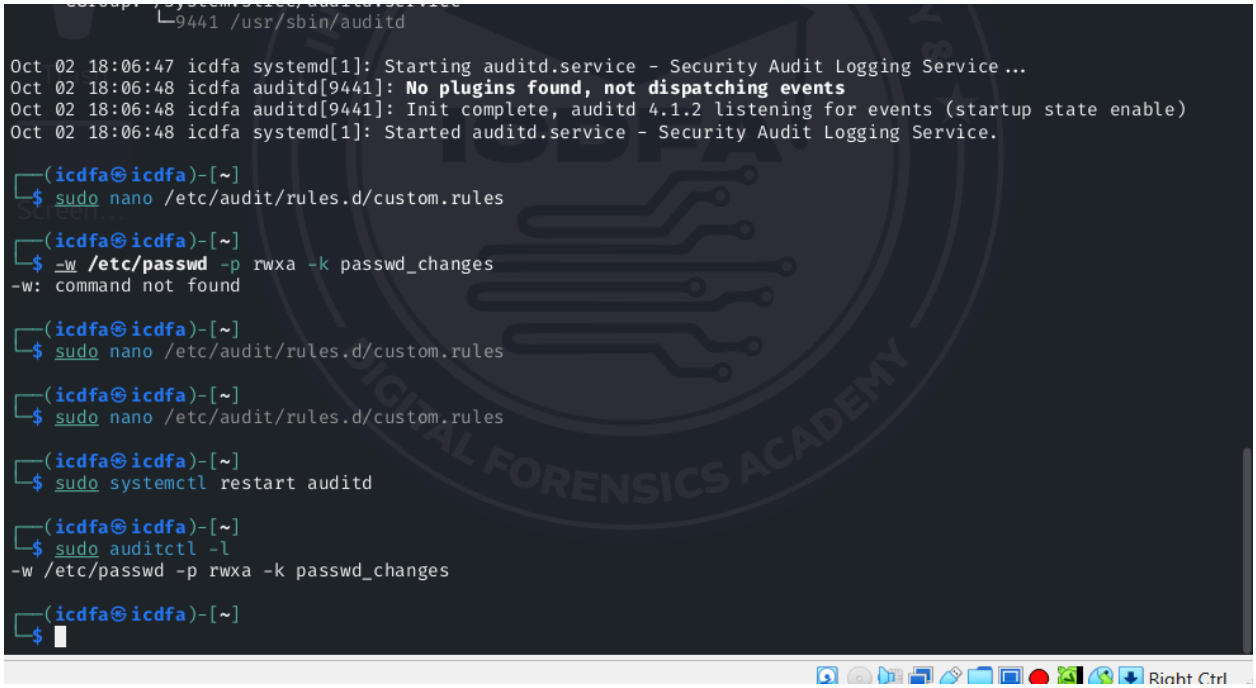

1 2 3 4
18:33
icdfa@icdfa: ~
File Actions Edit View Help
(icdfa㉿icdfa)-[~]
$ sudo journalctl
May 10 20:50:54 icdfa kernel: Linux version 6.12.13-amd64 (devel@kali.org) (x86_64-linux-gnu-gcc-14 (Debian 14>
May 10 20:50:54 icdfa kernel: Command line: BOOT_IMAGE=/boot/vmlinuz-6.12.13-amd64 root=UUID=a5089483-4c45-493>
May 10 20:50:54 icdfa kernel: BIOS-provided physical RAM map:
May 10 20:50:54 icdfa kernel: BIOS-e820: [mem 0x0000000000000000-0x000000000009fbff] usable
May 10 20:50:54 icdfa kernel: BIOS-e820: [mem 0x000000000009fc00-0x000000000009ffff] reserved
May 10 20:50:54 icdfa kernel: BIOS-e820: [mem 0x00000000000f0000-0x00000000000fffff] reserved
May 10 20:50:54 icdfa kernel: BIOS-e820: [mem 0x0000000000100000-0x000000007ffeffff] usable
May 10 20:50:54 icdfa kernel: BIOS-e820: [mem 0x000000007fff0000-0x000000007fffffff] ACPI data
May 10 20:50:54 icdfa kernel: BIOS-e820: [mem 0x00000000fec00000-0x00000000fec00fff] reserved
May 10 20:50:54 icdfa kernel: BIOS-e820: [mem 0x00000000fee00000-0x00000000fee00fff] reserved
May 10 20:50:54 icdfa kernel: BIOS-e820: [mem 0x00000000fffc0000-0x00000000ffffffff] reserved
May 10 20:50:54 icdfa kernel: NX (Execute Disable) protection: active
May 10 20:50:54 icdfa kernel: APIC: Static calls initialized
May 10 20:50:54 icdfa kernel: SMBIOS 2.5 present.
May 10 20:50:54 icdfa kernel: DMI: innotek GmbH VirtualBox/VirtualBox, BIOS VirtualBox 12/01/2006
May 10 20:50:54 icdfa kernel: DMI: Memory slots populated: 0/0
May 10 20:50:54 icdfa kernel: Hypervisor detected: KVM
May 10 20:50:54 icdfa kernel: kvm-clock: Using msrs 4b564d01 and 4b564d00
May 10 20:50:54 icdfa kernel: kvm-clock: using sched offset of 155397490851 cycles
May 10 20:50:54 icdfa kernel: clocksource: kvm-clock: mask: 0xffffffffffffffff max_cycles: 0x1cd42e4dffb, max_>
May 10 20:50:54 icdfa kernel: tsc: Detected 2419.196 MHz processor
May 10 20:50:54 icdfa kernel: e820: update [mem 0x00000000-0x00000fff] usable ==> reserved
May 10 20:50:54 icdfa kernel: e820: remove [mem 0x000a0000-0x000fffff] usable
May 10 20:50:54 icdfa kernel: last_pfn = 0x80000 max_arch_pfn = 0x400000000
May 10 20:50:54 icdfa kernel: MTRR map: 4 entries (3 fixed + 1 variable; max 35), built from 16 variable MTRRs
May 10 20:50:54 icdfa kernel: x86/PAT: Configuration [0-7]: WB WC UC- UC WB WP UC- WT
May 10 20:50:54 icdfa kernel: found SMP MP-table at [mem 0x0009fff0-0x0009ffff]
May 10 20:50:54 icdfa kernel: RAMDISK: [mem 0x282fb000-0x30174fff]
May 10 20:50:54 icdfa kernel: ACPI: Early table checksum verification disabled
May 10 20:50:54 icdfa kernel: ACPI: RSDP 0x00000000000E0000 000024 (v02 VBOX)
May 10 20:50:54 icdfa kernel: ACPI: XSDT 0x000000007FFF0030 00003C (v01 VBOX VBOXXSDT 00000001 ASL 00000061)
May 10 20:50:54 icdfa kernel: ACPI: FACP 0x000000007FFF00F0 0000F4 (v04 VBOX VBOXFACP 00000001 ASL 00000061)
May 10 20:50:54 icdfa kernel: ACPI: DSDT 0x000000007FFF0610 002353 (v02 VBOX VBOXBIOS 00000002 INTL 20100528)
May 10 20:50:54 icdfa kernel: ACPI: FACS 0x000000007FFF0200 000040
May 10 20:50:54 icdfa kernel: ACPI: FACS 0x000000007FFF0200 000040
May 10 20:50:54 icdfa kernel: ACPI: APIC 0x000000007FFF0240 000054 (v02 VBOX VBOXAPIC 00000001 ASL 00000061)
May 10 20:50:54 icdfa kernel: ACPI: SSDT 0x000000007FFF02A0 00036C (v01 VBOX VBOXCPUT 00000002 INTL 20100528)
May 10 20:50:54 icdfa kernel: ACPI: Reserving FACP table memory at [mem 0x7fff00f0-0x7fff01e3]
Right Ctrl


File Machine View Input Devices Help
1 2 3 4
18:34
icdfa@icdfa: ~
File Actions Edit View Help
┌──(icdfa㉿icdfa)-[~]
└─$ sudo journalctl --since "today"
Oct 02 17:54:03 icdfa kernel: Linux version 6.12.13-amd64 (devel@kali.org) (x86_64-linux-gnu-gcc-14 (Debian 14>
Oct 02 17:54:03 icdfa kernel: Command line: BOOT_IMAGE=/boot/vmlinuz-6.12.13-amd64 root=UUID=a5089483-4c45-493>
Oct 02 17:54:03 icdfa kernel: BIOS-provided physical RAM map:
Oct 02 17:54:03 icdfa kernel: BIOS-e820: [mem 0x0000000000000000-0x000000000009fbff] usable
Oct 02 17:54:03 icdfa kernel: BIOS-e820: [mem 0x000000000009fc00-0x000000000009ffff] reserved
Oct 02 17:54:03 icdfa kernel: BIOS-e820: [mem 0x00000000000f0000-0x00000000000fffff] reserved
Oct 02 17:54:03 icdfa kernel: BIOS-e820: [mem 0x0000000000100000-0x00000007ffefffff] usable
Oct 02 17:54:03 icdfa kernel: BIOS-e820: [mem 0x00000007fff0000-0x00000007ffffffff] ACPI data
Oct 02 17:54:03 icdfa kernel: BIOS-e820: [mem 0x00000000fec00000-0x00000000fec00fff] reserved
Oct 02 17:54:03 icdfa kernel: BIOS-e820: [mem 0x00000000fee00000-0x00000000fee00fff] reserved
Oct 02 17:54:03 icdfa kernel: BIOS-e820: [mem 0x00000000fffc0000-0x00000000ffffffff] reserved
Oct 02 17:54:03 icdfa kernel: NX (Execute Disable) protection: active
Oct 02 17:54:03 icdfa kernel: APIC: Static calls initialized
Oct 02 17:54:03 icdfa kernel: SMBIOS 2.5 present.
Oct 02 17:54:03 icdfa kernel: DMI: innotek GmbH VirtualBox/VirtualBox, BIOS VirtualBox 12/01/2006
Oct 02 17:54:03 icdfa kernel: DMI: Memory slots populated: 0/0
Oct 02 17:54:03 icdfa kernel: Hypervisor detected: KVM
Oct 02 17:54:03 icdfa kernel: kvm-clock: Using msrs 4b564d01 and 4b564d00
Oct 02 17:54:03 icdfa kernel: kvm-clock: using sched offset of 16387970006 cycles
Oct 02 17:54:03 icdfa kernel: clocksource: kvm-clock: mask: 0xffffffffffffffff max_cycles: 0x1cd42e4dffb, max_>
Oct 02 17:54:03 icdfa kernel: tsc: Detected 3110.412 MHz processor
Oct 02 17:54:03 icdfa kernel: e820: update [mem 0x00000000-0x00000fff] usable ==> reserved
Oct 02 17:54:03 icdfa kernel: e820: remove [mem 0x000a0000-0x000fffff] usable
Oct 02 17:54:03 icdfa kernel: last_pfn = 0x80000 max_arch_pfn = 0x400000000
Oct 02 17:54:03 icdfa kernel: MTRR map: 3 entries (3 fixed + 0 variable; max 19), built from 8 variable MTRRs
Oct 02 17:54:03 icdfa kernel: x86/PAT: Configuration [0-7]: WB WC UC- UC WB WP UC- WT
Oct 02 17:54:03 icdfa kernel: CPU MTRRs all blank - virtualized system.
Oct 02 17:54:03 icdfa kernel: found SMP MP-table at [mem 0x0009fbf0-0x0009fbff]
Oct 02 17:54:03 icdfa kernel: Incomplete global flushes, disabling PCID
Oct 02 17:54:03 icdfa kernel: RAMDISK: [mem 0x280ad000-0x3004dfff]
Oct 02 17:54:03 icdfa kernel: ACPI: Early table checksum verification disabled
Oct 02 17:54:03 icdfa kernel: ACPI: RSDP 0x00000000000E0000 000024 (v02 VBOX)
Oct 02 17:54:03 icdfa kernel: ACPI: XSDT 0x000000007FFF0030 00003C (v01 VBOX VBOXXSDT 00000001 ASL 00000061)
Oct 02 17:54:03 icdfa kernel: ACPI: FACP 0x000000007FFF00F0 0000F4 (v04 VBOX VBOXFACP 00000001 ASL 00000061)
Oct 02 17:54:03 icdfa kernel: ACPI: DSDT 0x000000007FFF02E0 002353 (v02 VBOX VBOXBIOS 00000002 VBOX 0002A935)
Oct 02 17:54:03 icdfa kernel: ACPI: FACS 0x000000007FFF0200 000040
Oct 02 17:54:03 icdfa kernel: ACPI: FACS 0x000000007FFF0200 000040
Right Ctrl


File Machine View Input Devices Help
1 2 3 4
18:34
icdfa@icdfa: ~
File Actions Edit View Help
┌──(icdfa㉿icdfa)-[~]
└─$ sudo journalctl -p err
May 10 20:50:54 icdfa kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* vmwgfx seems to be running on an unsupported
May 10 20:50:54 icdfa kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* This configuration is likely broken.
May 10 20:50:54 icdfa kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* Please switch to a supported graphics device
May 10 20:57:08 icdfa lightdm[761]: gkr-pam: unable to locate daemon control file
May 10 20:57:08 icdfa lightdm[645]: pam_systemd(lightdm-greeter:session): Failed to release session: Transport
May 10 20:57:21 icdfa obexd[1254]: Unable to acquire registery: Error calling StartServiceByName for org.gnome
May 10 20:57:21 icdfa obexd[1254]: Unable to acquire registery: Error calling StartServiceByName for org.gnome
-- Boot f8a18b2290a640c9b9236a0ff3f2a536 --
May 11 04:45:41 icdfa kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* vmwgfx seems to be running on an unsupported
May 11 04:45:41 icdfa kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* This configuration is likely broken.
May 11 04:45:41 icdfa kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* Please switch to a supported graphics device
May 11 04:51:38 icdfa lightdm[802]: gkr-pam: unable to locate daemon control file
May 11 04:51:38 icdfa lightdm[690]: pam_systemd(lightdm-greeter:session): Failed to release session: Transport
May 11 04:51:48 icdfa obexd[1324]: Unable to acquire registery: Error calling StartServiceByName for org.gnome
May 11 04:51:48 icdfa obexd[1324]: Unable to acquire registery: Error calling StartServiceByName for org.gnome
-- Boot 1330474cb531439e929342180494b74a --
May 11 06:10:01 icdfa kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* vmwgfx seems to be running on an unsupported
May 11 06:10:01 icdfa kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* This configuration is likely broken.
May 11 06:10:01 icdfa kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* Please switch to a supported graphics device
May 11 06:10:30 icdfa lightdm[785]: gkr-pam: unable to locate daemon control file
May 11 06:10:30 icdfa lightdm[678]: pam_systemd(lightdm-greeter:session): Failed to release session: Transport
May 11 06:10:42 icdfa obexd[1293]: Unable to acquire registery: Error calling StartServiceByName for org.gnome
May 11 06:10:42 icdfa obexd[1293]: Unable to acquire registery: Error calling StartServiceByName for org.gnome
-- Boot b8f421d015a94e7d82bf378d620f3ea1 --
May 11 06:22:39 icdfa kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* vmwgfx seems to be running on an unsupported
May 11 06:22:39 icdfa kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* This configuration is likely broken.
May 11 06:22:39 icdfa kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* Please switch to a supported graphics device
May 11 06:23:02 icdfa lightdm[741]: gkr-pam: unable to locate daemon control file
May 11 06:23:02 icdfa lightdm[635]: pam_systemd(lightdm-greeter:session): Failed to release session: Transport
May 11 06:23:17 icdfa obexd[1256]: Unable to acquire registery: Error calling StartServiceByName for org.gnome
May 11 06:23:17 icdfa obexd[1256]: Unable to acquire registery: Error calling StartServiceByName for org.gnome
-- Boot 70de93df416a4f15a4e11555f8166985 --
May 11 06:29:53 icdfa kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* vmwgfx seems to be running on an unsupported
May 11 06:29:53 icdfa kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* This configuration is likely broken.
May 11 06:29:53 icdfa kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* Please switch to a supported graphics device
May 11 06:30:16 icdfa lightdm[740]: gkr-pam: unable to locate daemon control file
May 11 06:30:16 icdfa lightdm[635]: pam_systemd(lightdm-greeter:session): Failed to release session: Transport
Right Ctrl


1 2 3 4
18:35
icdfa@icdfa: ~
File Actions Edit View Help
-- Boot 70de93df416a4f15a4e11555f8166985 --
May 11 06:29:53 icdfa kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* vmwgfx seems to be running on an unsupported
May 11 06:29:53 icdfa kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* This configuration is likely broken.
May 11 06:29:53 icdfa kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* Please switch to a supported graphics device
May 11 06:30:16 icdfa lightdm[740]: gkr-pam: unable to locate daemon control file
May 11 06:30:16 icdfa lightdm[635]: pam_systemd(lightdm-greeter:session): Failed to release session: Transport
-- Boot ed33344071284c619cc899cd9af2cfed --
May 11 06:32:09 icdfa kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* vmwgfx seems to be running on an unsupported
(icdfa㉿icdfa)-[~]
$ sudo journalctl -f
Oct 02 18:34:54 icdfa kernel: 17:34:54.213731 SHCLX11 Shared Clipboard: Converting VBox formats 'BITMAP' to '
INVALID' for X11 (idxFmtX11=0, fmtX11=0, atomT
Oct 02 18:34:54 icdfa kernel: 17:34:54.215109 SHCLX11 Shared Clipboard: Converting VBox formats 'BITMAP' to '
INVALID' for X11 (idxFmtX11=0, fmtX11=0, atomT
Oct 02 18:34:54 icdfa kernel: 17:34:54.509759 SHCLX11 Shared Clipboard: Converting VBox formats 'BITMAP' to '
INVALID' for X11 (idxFmtX11=0, fmtX11=0, atomT
Oct 02 18:34:54 icdfa kernel: 17:34:54.510562 SHCLX11 Shared Clipboard: Converting VBox formats 'BITMAP' to '
INVALID' for X11 (idxFmtX11=0, fmtX11=0, atomT
Oct 02 18:35:01 icdfa CRON[23858]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
Oct 02 18:35:01 icdfa CRON[23860]: (root) CMD (command -v debian-sa1 > /dev/null && debian-sa1 1 1)
Oct 02 18:35:02 icdfa CRON[23858]: pam_unix(cron:session): session closed for user root
Oct 02 18:35:19 icdfa sudo[23065]: pam_unix(sudo:session): session closed for user root
Oct 02 18:35:32 icdfa sudo[24119]: icdfa : TTY=pts/0 ; PWD=/home/icdfa ; USER=root ; COMMAND=/usr/bin/journa
lctl -f
Oct 02 18:35:32 icdfa sudo[24119]: pam_unix(sudo:session): session opened for user root(uid=0) by icdfa(uid=100
0)
Oct 02 18:35:44 icdfa kernel: 17:34:54.510813 SHCLX11 Shared Clipboard: Converting VBox formats 'BITMAP' to '
INVALID' for X11 (idxFmtX11=0, fmtX11=0, atomT
Oct 02 18:35:44 icdfa kernel: 17:35:44.936571 SHCLX11 Shared Clipboard: Converting VBox formats 'BITMAP' to '
INVALID' for X11 (idxFmtX11=0, fmtX11=0, atomT
Oct 02 18:35:44 icdfa kernel: 17:35:44.938165 SHCLX11 Shared Clipboard: Converting VBox formats 'BITMAP' to '
INVALID' for X11 (idxFmtX11=0, fmtX11=0, atomT
Oct 02 18:35:45 icdfa kernel: 17:35:44.939747 SHCLX11 Shared Clipboard: Converting VBox formats 'BITMAP' to '
INVALID' for X11 (idxFmtX11=0, fmtX11=0, atomT
Oct 02 18:35:45 icdfa kernel: 17:35:45.227104 SHCLX11 Shared Clipboard: Converting VBox formats 'BITMAP' to '
INVALID' for X11 (idxFmtX11=0, fmtX11=0, atomT
Oct 02 18:35:45 icdfa kernel: 17:35:45.228061 SHCLX11 Shared Clipboard: Converting VBox formats 'BITMAP' to '
INVALID' for X11 (idxFmtX11=0, fmtX11=0, atomT
Right Ctrl


1 2 3 4
18:52
icdfa@icdfa: ~
File Actions Edit View Help
(icdfa㉿icdfa)-[~]
$ sudo lynis audit system
[Lynis 3.1.4]
Lynis comes with ABSOLUTELY NO WARRANTY. This is free software, and you are
welcome to redistribute it under the terms of the GNU General Public License.
See the LICENSE file for details about using this software.
2007-2024, CISOfy - https://cisofy.com/lynis/
Enterprise support available (compliance, plugins, interface and tools)
[+] Initializing program
- Detecting OS... [DONE]
- Checking profiles... [DONE]
Program version: 3.1.4
Operating system: Linux
Operating system name: Kali Linux
Operating system version: Rolling release
Kernel version: 6.12.13
Hardware platform: x86_64
Hostname: icdfa
Profiles: /etc/lynis/default.prf
Log file: /var/log/lynis.log
Report file: /var/log/lynis-report.dat
Report version: 1.0
Plugin directory: /etc/lynis/plugins
Auditor: [Not Specified]
Language: en
Test category: all
Test group: all
- Program update status... [NO UPDATE]
[+] System tools
- Scanning available tools...
- Checking system binaries...
Right Ctrl



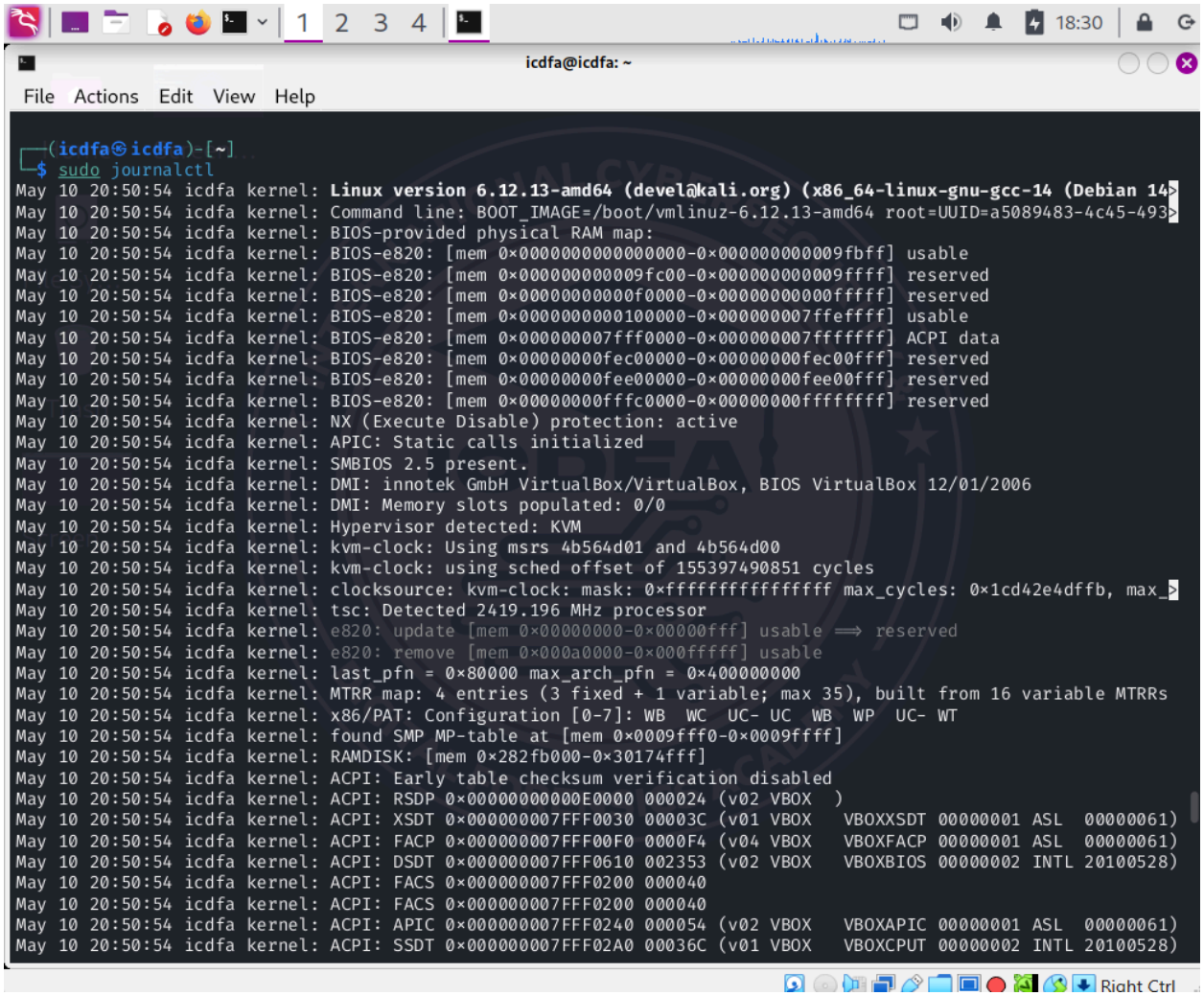
Evidence source	How it reaches the SIEM	Technical alert (SOC)	Governance issue
auditd (`passwd_changes`, `shadow_changes`, `execve`)	Forwarding agent (e.g. Wazuh, Filebeat, Splunk UF) or rsyslog	Change to `/etc/shadow` outside a change window	Repeated unapproved changes; no change ticket linked
auth.log / journald	Syslog or agent collection	Burst of failed logins from one source; unexpected sudo use	Persistent failures over time; privileged use without approval
Lynis report	Scheduled scan, `lynis-report.dat` parsed and ingested	Single critical warning	Hardening index trending down; findings past remediation SLA
Service health (auditd, rsyslog)	Heartbeat / health check	auditd stopped or rules not loaded	Logging gap beyond tolerance on a critical system

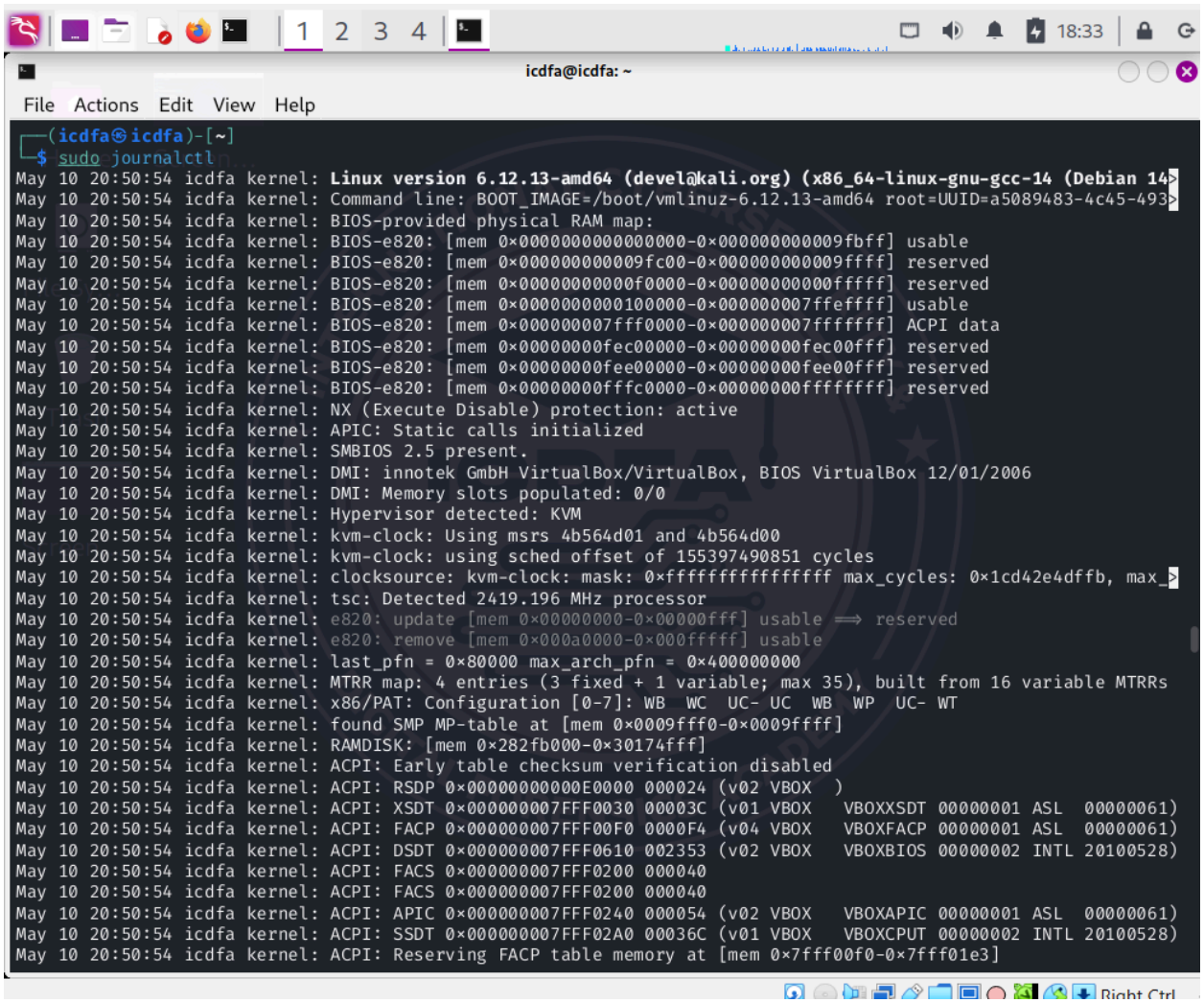
Suggested escalation thresholds (adjust to your organisation's policy):

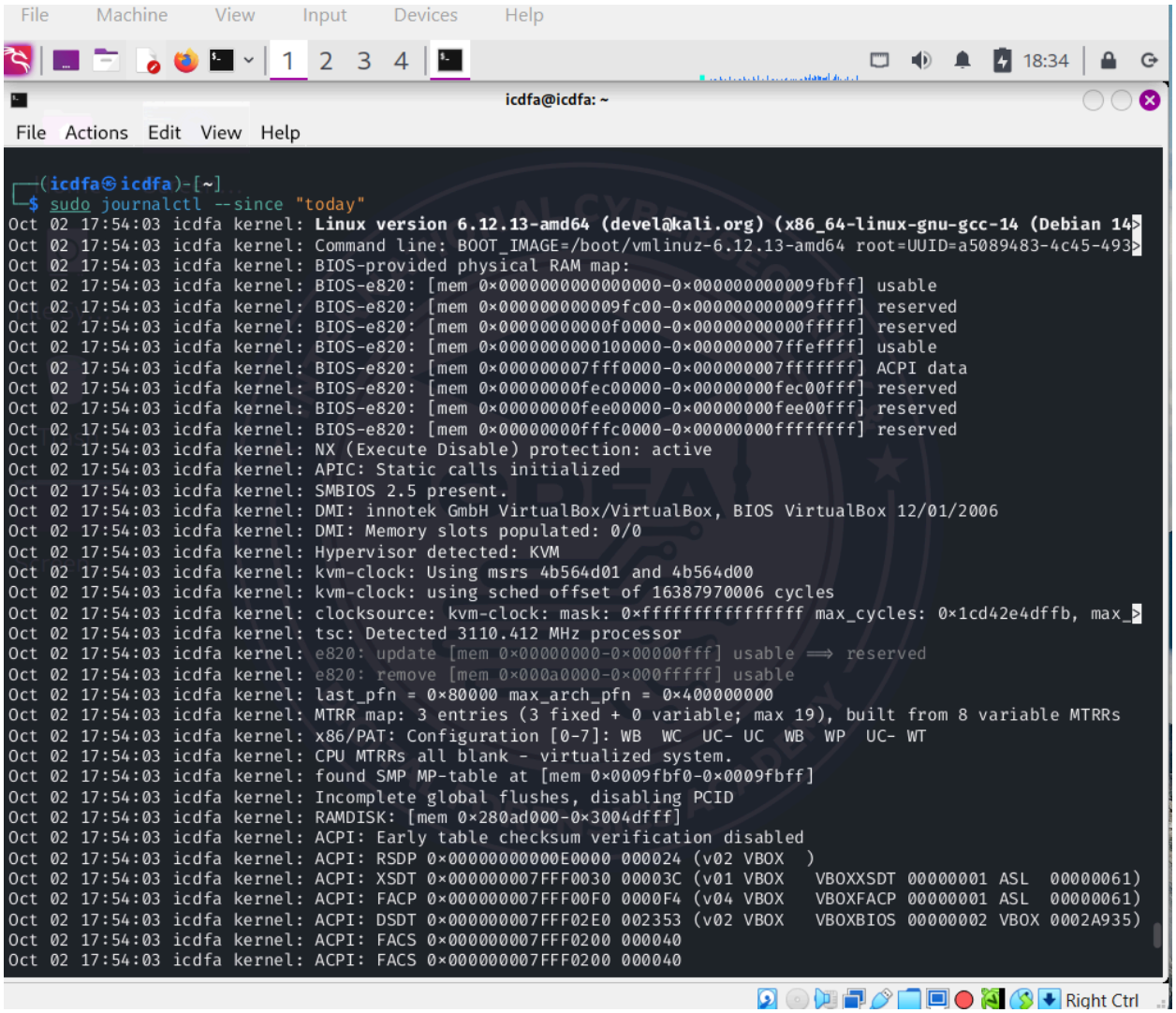
- **CISO:** audit logging disabled or a gap of more than 24 hours on a critical system; unauthorised change to account files; a critical finding unremediated beyond its SLA.
- **Risk function:** repeated overdue high findings, or a hardening index below the approved baseline.
- **Management committee:** the same KRI breached in consecutive periods, or evidence collection failing across multiple systems (which matches the scenario's "inconsistent evidence").

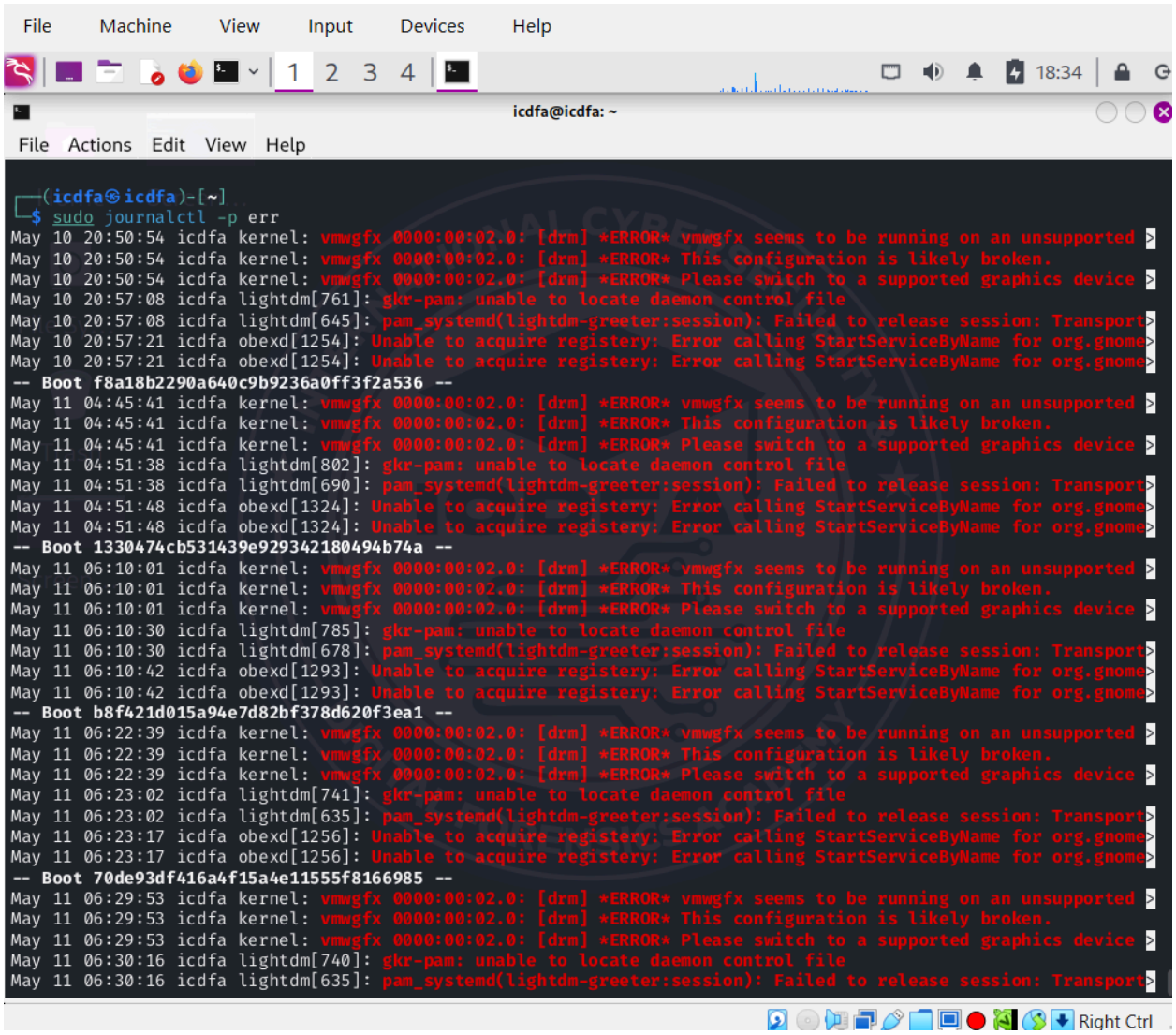
Generic answers to the escalation questions: closure evidence is a re-run of the same command or Lynis check showing the corrected state, plus a change ticket. Retest after remediation, then on a recurring schedule such as monthly for high findings, driven by automated scans.

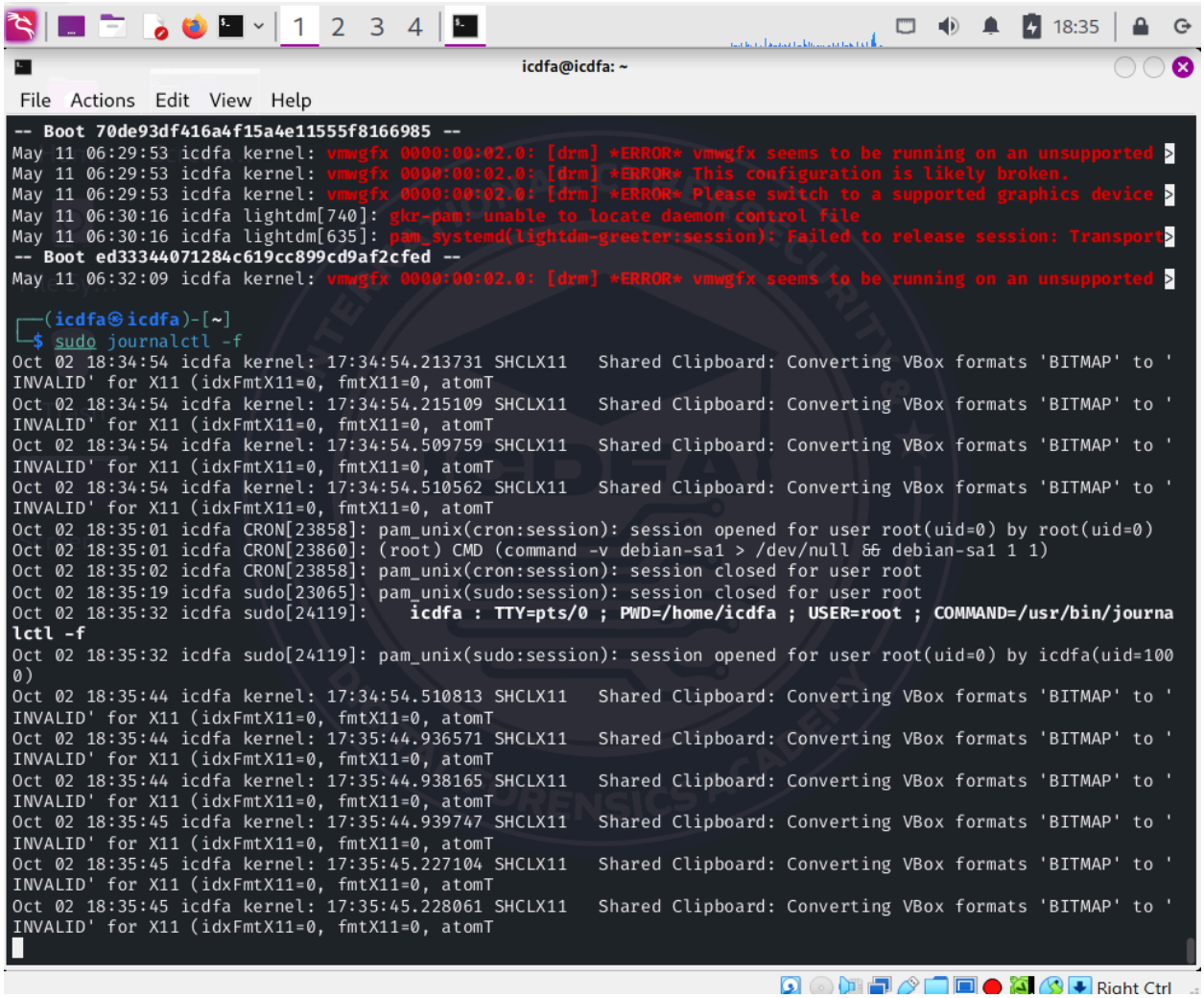
ACTIVITY 2


1 2 3 4
18:30
icdfa@icdfa: ~
File Actions Edit View Help
┌──(icdfa㉿icdfa)-[~]
└─$ sudo journalctl
May 10 20:50:54 icdfa kernel: Linux version 6.12.13-amd64 (devel@kali.org) (x86_64-linux-gnu-gcc-14 (Debian 14>
May 10 20:50:54 icdfa kernel: Command line: BOOT_IMAGE=/boot/vmlinuz-6.12.13-amd64 root=UUID=a5089483-4c45-493>
May 10 20:50:54 icdfa kernel: BIOS-provided physical RAM map:
May 10 20:50:54 icdfa kernel: BIOS-e820: [mem 0x0000000000000000-0x000000000009fbff] usable
May 10 20:50:54 icdfa kernel: BIOS-e820: [mem 0x000000000009fc00-0x000000000009ffff] reserved
May 10 20:50:54 icdfa kernel: BIOS-e820: [mem 0x00000000000f0000-0x00000000000fffff] reserved
May 10 20:50:54 icdfa kernel: BIOS-e820: [mem 0x0000000000100000-0x000000007ffeffff] usable
May 10 20:50:54 icdfa kernel: BIOS-e820: [mem 0x000000007fff0000-0x000000007fffffff] ACPI data
May 10 20:50:54 icdfa kernel: BIOS-e820: [mem 0x00000000fec00000-0x00000000fec00fff] reserved
May 10 20:50:54 icdfa kernel: BIOS-e820: [mem 0x00000000fee00000-0x00000000fee00fff] reserved
May 10 20:50:54 icdfa kernel: BIOS-e820: [mem 0x00000000fffc0000-0x00000000ffffffff] reserved
May 10 20:50:54 icdfa kernel: NX (Execute Disable) protection: active
May 10 20:50:54 icdfa kernel: APIC: Static calls initialized
May 10 20:50:54 icdfa kernel: SMBIOS 2.5 present.
May 10 20:50:54 icdfa kernel: DMI: innotek GmbH VirtualBox/VirtualBox, BIOS VirtualBox 12/01/2006
May 10 20:50:54 icdfa kernel: DMI: Memory slots populated: 0/0
May 10 20:50:54 icdfa kernel: Hypervisor detected: KVM
May 10 20:50:54 icdfa kernel: kvm-clock: Using msrs 4b564d01 and 4b564d00
May 10 20:50:54 icdfa kernel: kvm-clock: using sched offset of 155397490851 cycles
May 10 20:50:54 icdfa kernel: clocksource: kvm-clock: mask: 0xffffffffffffffff max_cycles: 0x1cd42e4dffb, max_>
May 10 20:50:54 icdfa kernel: tsc: Detected 2419.196 MHz processor
May 10 20:50:54 icdfa kernel: e820: update [mem 0x00000000-0x00000fff] usable ==> reserved
May 10 20:50:54 icdfa kernel: e820: remove [mem 0x000a0000-0x000fffff] usable
May 10 20:50:54 icdfa kernel: last_pfn = 0x80000 max_arch_pfn = 0x400000000
May 10 20:50:54 icdfa kernel: MTRR map: 4 entries (3 fixed + 1 variable; max 35), built from 16 variable MTRRs
May 10 20:50:54 icdfa kernel: x86/PAT: Configuration [0-7]: WB WC UC- UC WB WP UC- WT
May 10 20:50:54 icdfa kernel: found SMP MP-table at [mem 0x0009fff0-0x0009ffff]
May 10 20:50:54 icdfa kernel: RAMDISK: [mem 0x282fb000-0x30174fff]
May 10 20:50:54 icdfa kernel: ACPI: Early table checksum verification disabled
May 10 20:50:54 icdfa kernel: ACPI: RSDP 0x00000000000E0000 000024 (v02 VBOX)
May 10 20:50:54 icdfa kernel: ACPI: XSDT 0x000000007FFF0030 00003C (v01 VBOX VBOXXSDT 00000001 ASL 00000061)
May 10 20:50:54 icdfa kernel: ACPI: FACP 0x000000007FFF00F0 0000F4 (v04 VBOX VBOXFACP 00000001 ASL 00000061)
May 10 20:50:54 icdfa kernel: ACPI: DSDT 0x000000007FFF0610 002353 (v02 VBOX VBOXBIOS 00000002 INTL 20100528)
May 10 20:50:54 icdfa kernel: ACPI: FACS 0x000000007FFF0200 000040
May 10 20:50:54 icdfa kernel: ACPI: FACS 0x000000007FFF0200 000040
May 10 20:50:54 icdfa kernel: ACPI: APIC 0x000000007FFF0240 000054 (v02 VBOX VBOXAPIC 00000001 ASL 00000061)
May 10 20:50:54 icdfa kernel: ACPI: SSDT 0x000000007FFF02A0 00036C (v01 VBOX VBOXCPUT 00000002 INTL 20100528)
Right Ctrl


1 2 3 4
18:33
icdfa@icdfa: ~
File Actions Edit View Help
(icdfa㉿icdfa)-[~]
$ sudo journalctl
May 10 20:50:54 icdfa kernel: Linux version 6.12.13-amd64 (devel@kali.org) (x86_64-linux-gnu-gcc-14 (Debian 14>
May 10 20:50:54 icdfa kernel: Command line: BOOT_IMAGE=/boot/vmlinuz-6.12.13-amd64 root=UUID=a5089483-4c45-493>
May 10 20:50:54 icdfa kernel: BIOS-provided physical RAM map:
May 10 20:50:54 icdfa kernel: BIOS-e820: [mem 0x0000000000000000-0x000000000009fbff] usable
May 10 20:50:54 icdfa kernel: BIOS-e820: [mem 0x000000000009fc00-0x000000000009ffff] reserved
May 10 20:50:54 icdfa kernel: BIOS-e820: [mem 0x00000000000f0000-0x00000000000fffff] reserved
May 10 20:50:54 icdfa kernel: BIOS-e820: [mem 0x0000000000100000-0x000000007ffeffff] usable
May 10 20:50:54 icdfa kernel: BIOS-e820: [mem 0x000000007fff0000-0x000000007fffffff] ACPI data
May 10 20:50:54 icdfa kernel: BIOS-e820: [mem 0x00000000fec00000-0x00000000fec00fff] reserved
May 10 20:50:54 icdfa kernel: BIOS-e820: [mem 0x00000000fee00000-0x00000000fee00fff] reserved
May 10 20:50:54 icdfa kernel: BIOS-e820: [mem 0x00000000fffc0000-0x00000000ffffffff] reserved
May 10 20:50:54 icdfa kernel: NX (Execute Disable) protection: active
May 10 20:50:54 icdfa kernel: APIC: Static calls initialized
May 10 20:50:54 icdfa kernel: SMBIOS 2.5 present.
May 10 20:50:54 icdfa kernel: DMI: innotek GmbH VirtualBox/VirtualBox, BIOS VirtualBox 12/01/2006
May 10 20:50:54 icdfa kernel: DMI: Memory slots populated: 0/0
May 10 20:50:54 icdfa kernel: Hypervisor detected: KVM
May 10 20:50:54 icdfa kernel: kvm-clock: Using msrs 4b564d01 and 4b564d00
May 10 20:50:54 icdfa kernel: kvm-clock: using sched offset of 155397490851 cycles
May 10 20:50:54 icdfa kernel: clocksource: kvm-clock: mask: 0xffffffffffffffff max_cycles: 0x1cd42e4dffb, max_>
May 10 20:50:54 icdfa kernel: tsc: Detected 2419.196 MHz processor
May 10 20:50:54 icdfa kernel: e820: update [mem 0x00000000-0x00000fff] usable ==> reserved
May 10 20:50:54 icdfa kernel: e820: remove [mem 0x000a0000-0x000fffff] usable
May 10 20:50:54 icdfa kernel: last_pfn = 0x80000 max_arch_pfn = 0x400000000
May 10 20:50:54 icdfa kernel: MTRR map: 4 entries (3 fixed + 1 variable; max 35), built from 16 variable MTRRs
May 10 20:50:54 icdfa kernel: x86/PAT: Configuration [0-7]: WB WC UC- UC WB WP UC- WT
May 10 20:50:54 icdfa kernel: found SMP MP-table at [mem 0x0009fff0-0x0009ffff]
May 10 20:50:54 icdfa kernel: RAMDISK: [mem 0x282fb000-0x30174fff]
May 10 20:50:54 icdfa kernel: ACPI: Early table checksum verification disabled
May 10 20:50:54 icdfa kernel: ACPI: RSDP 0x00000000000E0000 000024 (v02 VBOX)
May 10 20:50:54 icdfa kernel: ACPI: XSDT 0x000000007FFF0030 00003C (v01 VBOX VBOXXSDT 00000001 ASL 00000061)
May 10 20:50:54 icdfa kernel: ACPI: FACP 0x000000007FFF00F0 0000F4 (v04 VBOX VBOXFACP 00000001 ASL 00000061)
May 10 20:50:54 icdfa kernel: ACPI: DSDT 0x000000007FFF0610 002353 (v02 VBOX VBOXBIOS 00000002 INTL 20100528)
May 10 20:50:54 icdfa kernel: ACPI: FACS 0x000000007FFF0200 000040
May 10 20:50:54 icdfa kernel: ACPI: FACS 0x000000007FFF0200 000040
May 10 20:50:54 icdfa kernel: ACPI: APIC 0x000000007FFF0240 000054 (v02 VBOX VBOXAPIC 00000001 ASL 00000061)
May 10 20:50:54 icdfa kernel: ACPI: SSDT 0x000000007FFF02A0 00036C (v01 VBOX VBOXCPUT 00000002 INTL 20100528)
May 10 20:50:54 icdfa kernel: ACPI: Reserving FACP table memory at [mem 0x7fff00f0-0x7fff01e3]
Right Ctrl


File Machine View Input Devices Help
1 2 3 4
18:34
icdfa@icdfa: ~
File Actions Edit View Help
┌──(icdfa㉿icdfa)-[~]
└─$ sudo journalctl --since "today"
Oct 02 17:54:03 icdfa kernel: Linux version 6.12.13-amd64 (devel@kali.org) (x86_64-linux-gnu-gcc-14 (Debian 14>
Oct 02 17:54:03 icdfa kernel: Command line: BOOT_IMAGE=/boot/vmlinuz-6.12.13-amd64 root=UUID=a5089483-4c45-493>
Oct 02 17:54:03 icdfa kernel: BIOS-provided physical RAM map:
Oct 02 17:54:03 icdfa kernel: BIOS-e820: [mem 0x0000000000000000-0x000000000009fbff] usable
Oct 02 17:54:03 icdfa kernel: BIOS-e820: [mem 0x000000000009fc00-0x000000000009ffff] reserved
Oct 02 17:54:03 icdfa kernel: BIOS-e820: [mem 0x00000000000f0000-0x00000000000fffff] reserved
Oct 02 17:54:03 icdfa kernel: BIOS-e820: [mem 0x0000000000100000-0x00000007ffefffff] usable
Oct 02 17:54:03 icdfa kernel: BIOS-e820: [mem 0x00000007fff0000-0x00000007ffffffff] ACPI data
Oct 02 17:54:03 icdfa kernel: BIOS-e820: [mem 0x00000000fec00000-0x00000000fec00fff] reserved
Oct 02 17:54:03 icdfa kernel: BIOS-e820: [mem 0x00000000fee00000-0x00000000fee00fff] reserved
Oct 02 17:54:03 icdfa kernel: BIOS-e820: [mem 0x00000000fffc0000-0x00000000ffffffff] reserved
Oct 02 17:54:03 icdfa kernel: NX (Execute Disable) protection: active
Oct 02 17:54:03 icdfa kernel: APIC: Static calls initialized
Oct 02 17:54:03 icdfa kernel: SMBIOS 2.5 present.
Oct 02 17:54:03 icdfa kernel: DMI: innotek GmbH VirtualBox/VirtualBox, BIOS VirtualBox 12/01/2006
Oct 02 17:54:03 icdfa kernel: DMI: Memory slots populated: 0/0
Oct 02 17:54:03 icdfa kernel: Hypervisor detected: KVM
Oct 02 17:54:03 icdfa kernel: kvm-clock: Using msrs 4b564d01 and 4b564d00
Oct 02 17:54:03 icdfa kernel: kvm-clock: using sched offset of 16387970006 cycles
Oct 02 17:54:03 icdfa kernel: clocksource: kvm-clock: mask: 0xffffffffffffffff max_cycles: 0x1cd42e4dffb, max_>
Oct 02 17:54:03 icdfa kernel: tsc: Detected 3110.412 MHz processor
Oct 02 17:54:03 icdfa kernel: e820: update [mem 0x00000000-0x00000fff] usable ==> reserved
Oct 02 17:54:03 icdfa kernel: e820: remove [mem 0x000a0000-0x000fffff] usable
Oct 02 17:54:03 icdfa kernel: last_pfn = 0x80000 max_arch_pfn = 0x400000000
Oct 02 17:54:03 icdfa kernel: MTRR map: 3 entries (3 fixed + 0 variable; max 19), built from 8 variable MTRRs
Oct 02 17:54:03 icdfa kernel: x86/PAT: Configuration [0-7]: WB WC UC- UC WB WP UC- WT
Oct 02 17:54:03 icdfa kernel: CPU MTRRs all blank - virtualized system.
Oct 02 17:54:03 icdfa kernel: found SMP MP-table at [mem 0x0009fbf0-0x0009fbff]
Oct 02 17:54:03 icdfa kernel: Incomplete global flushes, disabling PCID
Oct 02 17:54:03 icdfa kernel: RAMDISK: [mem 0x280ad000-0x3004dfff]
Oct 02 17:54:03 icdfa kernel: ACPI: Early table checksum verification disabled
Oct 02 17:54:03 icdfa kernel: ACPI: RSDP 0x00000000000E0000 000024 (v02 VBOX)
Oct 02 17:54:03 icdfa kernel: ACPI: XSDT 0x000000007FFF0030 00003C (v01 VBOX VBOXXSDT 00000001 ASL 00000061)
Oct 02 17:54:03 icdfa kernel: ACPI: FACP 0x000000007FFF00F0 0000F4 (v04 VBOX VBOXFACP 00000001 ASL 00000061)
Oct 02 17:54:03 icdfa kernel: ACPI: DSDT 0x000000007FFF02E0 002353 (v02 VBOX VBOXBIOS 00000002 VBOX 0002A935)
Oct 02 17:54:03 icdfa kernel: ACPI: FACS 0x000000007FFF0200 000040
Oct 02 17:54:03 icdfa kernel: ACPI: FACS 0x000000007FFF0200 000040
Right Ctrl


File Machine View Input Devices Help
1 2 3 4
18:34
icdfa@icdfa: ~
File Actions Edit View Help
┌──(icdfa㉿icdfa)-[~]
└─$ sudo journalctl -p err
May 10 20:50:54 icdfa kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* vmwgfx seems to be running on an unsupported
May 10 20:50:54 icdfa kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* This configuration is likely broken.
May 10 20:50:54 icdfa kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* Please switch to a supported graphics device
May 10 20:57:08 icdfa lightdm[761]: gkr-pam: unable to locate daemon control file
May 10 20:57:08 icdfa lightdm[645]: pam_systemd(lightdm-greeter:session): Failed to release session: Transport
May 10 20:57:21 icdfa obexd[1254]: Unable to acquire registery: Error calling StartServiceByName for org.gnome
May 10 20:57:21 icdfa obexd[1254]: Unable to acquire registery: Error calling StartServiceByName for org.gnome
-- Boot f8a18b2290a640c9b9236a0ff3f2a536 --
May 11 04:45:41 icdfa kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* vmwgfx seems to be running on an unsupported
May 11 04:45:41 icdfa kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* This configuration is likely broken.
May 11 04:45:41 icdfa kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* Please switch to a supported graphics device
May 11 04:51:38 icdfa lightdm[802]: gkr-pam: unable to locate daemon control file
May 11 04:51:38 icdfa lightdm[690]: pam_systemd(lightdm-greeter:session): Failed to release session: Transport
May 11 04:51:48 icdfa obexd[1324]: Unable to acquire registery: Error calling StartServiceByName for org.gnome
May 11 04:51:48 icdfa obexd[1324]: Unable to acquire registery: Error calling StartServiceByName for org.gnome
-- Boot 1330474cb531439e929342180494b74a --
May 11 06:10:01 icdfa kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* vmwgfx seems to be running on an unsupported
May 11 06:10:01 icdfa kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* This configuration is likely broken.
May 11 06:10:01 icdfa kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* Please switch to a supported graphics device
May 11 06:10:30 icdfa lightdm[785]: gkr-pam: unable to locate daemon control file
May 11 06:10:30 icdfa lightdm[678]: pam_systemd(lightdm-greeter:session): Failed to release session: Transport
May 11 06:10:42 icdfa obexd[1293]: Unable to acquire registery: Error calling StartServiceByName for org.gnome
May 11 06:10:42 icdfa obexd[1293]: Unable to acquire registery: Error calling StartServiceByName for org.gnome
-- Boot b8f421d015a94e7d82bf378d620f3ea1 --
May 11 06:22:39 icdfa kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* vmwgfx seems to be running on an unsupported
May 11 06:22:39 icdfa kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* This configuration is likely broken.
May 11 06:22:39 icdfa kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* Please switch to a supported graphics device
May 11 06:23:02 icdfa lightdm[741]: gkr-pam: unable to locate daemon control file
May 11 06:23:02 icdfa lightdm[635]: pam_systemd(lightdm-greeter:session): Failed to release session: Transport
May 11 06:23:17 icdfa obexd[1256]: Unable to acquire registery: Error calling StartServiceByName for org.gnome
May 11 06:23:17 icdfa obexd[1256]: Unable to acquire registery: Error calling StartServiceByName for org.gnome
-- Boot 70de93df416a4f15a4e11555f8166985 --
May 11 06:29:53 icdfa kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* vmwgfx seems to be running on an unsupported
May 11 06:29:53 icdfa kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* This configuration is likely broken.
May 11 06:29:53 icdfa kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* Please switch to a supported graphics device
May 11 06:30:16 icdfa lightdm[740]: gkr-pam: unable to locate daemon control file
May 11 06:30:16 icdfa lightdm[635]: pam_systemd(lightdm-greeter:session): Failed to release session: Transport
Right Ctrl



Error/warning evidence (journald, via `journalctl -p err` and `journalctl -f`)

- **Graphics driver, kernel.** Logged at every boot, starting May 10 20:50:54: `vmwgfx seems to be running on an unsupported hardware`, `This configuration is likely broken`, `Please switch to a supported graphics device`.
- **Keyring/PAM, lightdm.** `lightdm[761]: gkr-pam: unable to locate daemon control file`, first seen May 10 20:57:08.
- **Session cleanup, lightdm-greeter.** `pam_systemd(lightdm-greeter:session): Failed to release session: Transport...` (message truncated on screen).
- **Bluetooth OBEX service, obexd.** `Unable to acquire registry: Error calling StartServiceByName for org.gnome...`, first seen May 10 20:57:21.
- **Warning, kernel/VBox.** `SHCLX11 Shared Clipboard: Converting VBox formats 'BITMAP' to 'INVALID' for X11`, repeating on Oct 2 at 18:34:54 and 18:35:44.
- **Informational, security-relevant.** `sudo[24119]: icdfa : TTY=pts/0 ; PWD=/home/icdfa ; USER=root ; COMMAND=/usr/bin/journalctl -f` at Oct 2 18:35:32, followed by a

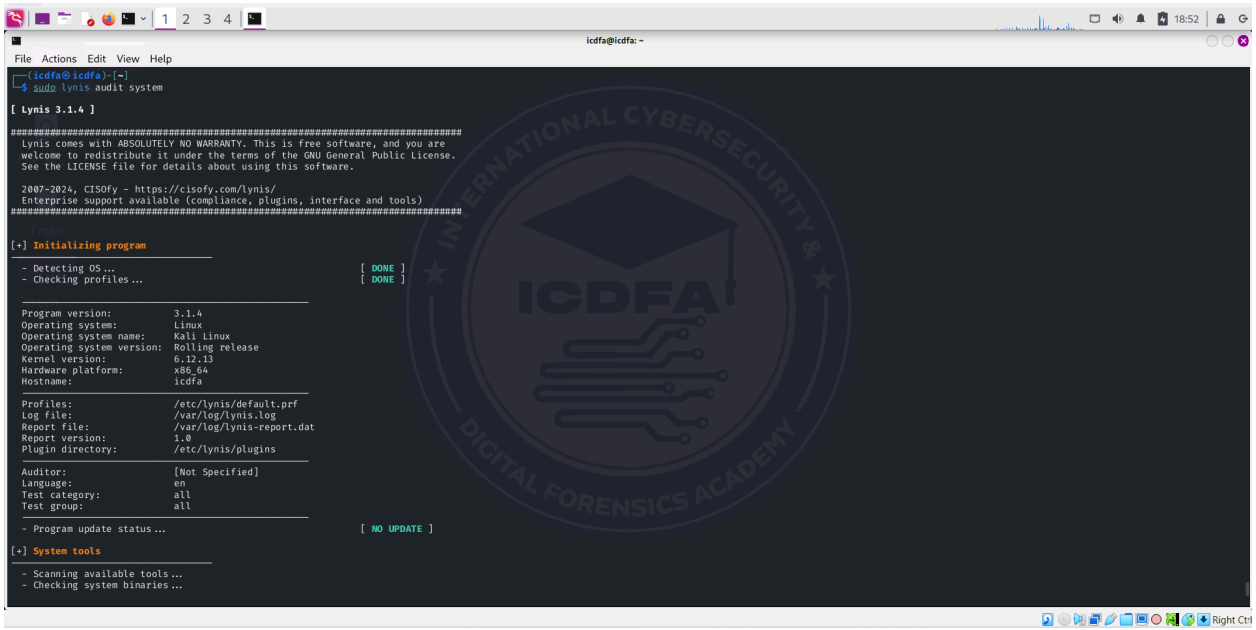
pam_unix(sudo:session) session opened for root by icdfa(uid=1000). A cron job also ran as root at 18:35:01: CMD (command -v debian-sa1 ...).

Event table

Event / Condition	Evidence Source	Time	User / Service	Security Significance	Recommended Action
vmwgfx driver on unsupported hardware, "configuration likely broken"	`journalctl -p err` (kernel)	May 10 20:50:54, repeated on every later boot (May 11 04:45, 06:10, 06:22, 06:29, 06:32)	kernel / vmwgfx	Low. It is a VirtualBox VM loading a VMware graphics driver. It is a misconfiguration, not an attack, but persistent errors create noise that can hide real events.	Set the VM graphics controller to VMSVGA or match the guest driver to the hypervisor. Then confirm the error no longer appears.
gkr-pam: unable to locate daemon control file	`journalctl -p err`	May 10 20:57:08, then each boot	lightdm (PAM)	Low to medium. A PAM module in the login path is failing, so the keyring does not unlock at login. Any broken authentication-path component deserves review.	Check the PAM/gnome-keyring configuration for the display manager. Confirm login still enforces authentication correctly.
Failed to release session: Transport...	`journalctl -p err`	May 10 20:57:08 and each boot	lightdm-greeter / pam_systemd	Low to medium. Failed session teardown could leave stale sessions, which weakens session accountability.	Review `loginctl list-sessions` for orphaned sessions. Update lightdm and systemd packages.
obexd cannot acquire registry (StartServiceByName)	`journalctl -p err`	May 10 20:57:21, May 11 04:51:48, 06:10:42, 06:23:17	obexd (Bluetooth OBEX)	Low, but it signals an unneeded service running on a lab VM. This enlarges the attack surface.	Disable or mask unused services (e.g. `systemctl disable --now obex`) to follow least functionality.
VBox shared clipboard conversion warnings (`BITMAP` to `INVALID`)	`journalctl -f` (kernel/SHCLX11)	Oct 2 18:34:54 to 18:35:45	VBox Guest Additions	Low in function. The shared clipboard is a host/guest data channel, so it is a potential data-leakage path.	Disable bidirectional clipboard in the VM settings unless it is needed.
Privileged command `sudo journalctl -f`	`journalctl -f` (sudo, pam_unix)	Oct 2 18:35:32	`icdfa` (uid 1000) to root	Informational. This is good evidence of privileged-use accountability, since user, TTY, directory, command and time are all recorded.	Retain these logs and forward them to central logging. Review sudo use regularly.
Scheduled root cron job (sysstat `debian-sa1`)	`journalctl -f` (CRON)	Oct 2 18:35:01 to 18:35:02	root via CRON	Informational, expected system activity. A root cron entry should still be known and baselined.	Confirm the job belongs to the sysstat package. Alert on new or unexpected cron entries.

Overall interpretation: the logs show no failed logins or intrusion indicators in these screenshots. The errors are recurring configuration and service faults. They matter for governance because repeated, unresolved errors normalise noise, unnecessary services widen the attack surface, and the sudo and cron records show the logging does support accountability. Journald also retained logs across several boots since May 10, which supports retention requirements.

ACTIVITY 3


icdfa@icdfa: ~
File Actions Edit View Help
(icdfa㉿icdfa)-[~]
$ sudo lynis audit system
[Lynis 3.1.4]
Lynis comes with ABSOLUTELY NO WARRANTY. This is free software, and you are
welcome to redistribute it under the terms of the GNU General Public License.
See the LICENSE file for details about using this software.
2007-2024, CISOfy - https://cisofy.com/lynis/
Enterprise support available (compliance, plugins, interface and tools)
[+] Initializing program
- Detecting OS... [DONE]
- Checking profiles... [DONE]
Program version: 3.1.4
Operating system: Linux
Operating system name: Kali Linux
Operating system version: Rolling release
Kernel version: 6.12.13
Hardware platform: x86_64
Hostname: icdfa
Profiles: /etc/lynis/default.prf
Log file: /var/log/lynis.log
Report file: /var/log/lynis-report.dat
Report version: 1.0
Plugin directory: /etc/lynis/plugins
Auditor: [Not Specified]
Language: en
Test category: all
Test group: all
- Program update status... [NO UPDATE]
[+] System tools
- Scanning available tools...
- Checking system binaries...
INTERNATIONAL CYBERSECURITY & DIGITAL FORENSICS ACADEMY
ICDFA
Right Ctrl

ACTIVITY 4

SIEM mapping

Evidence source	How it reaches the SIEM	Technical alert (SOC)	Governance issue
auditd (`passwd_changes`, `shadow_changes`, `execve`)	Forwarding agent (e.g. Wazuh, Filebeat, Splunk UF) or rsyslog	Change to `/etc/shadow` outside a change window	Repeated unapproved changes; no change ticket linked
auth.log / journald	Syslog or agent collection	Burst of failed logins from one source; unexpected sudo use	Persistent failures over time; privileged use without approval
Lynis report	Scheduled scan, `lynis-report.dat` parsed and ingested	Single critical warning	Hardening index trending down; findings past remediation SLA
Service health (auditd, rsyslog)	Heartbeat / health check	auditd stopped or rules not loaded	Logging gap beyond tolerance on a critical system

Suggested escalation thresholds:

- **CISO:** audit logging disabled or a gap of more than 24 hours on a critical system; unauthorised change to account files; a critical finding unremediated beyond its SLA.
- **Risk function:** repeated overdue high findings, or a hardening index below the approved baseline.
- **Management committee:** the same KRI breached in consecutive periods, or evidence collection failing across multiple systems (which matches the scenario's "inconsistent evidence").

Using the evidence collected in Modules 1–3, identify at least three events, weaknesses or conditions that warrant governance attention. These may include repeated authentication failures, unexpected privileged activity, missing audit coverage, overdue hardening issues, inadequate logging, critical Lynis findings or another evidence-based condition.

Control-Monitoring Table

Control / Objective	Evidence	Owner	Observed Status	KPI/KRI or Threshold	Risk / Significance	Remediation	Retest / Follow-up
1. Audit rule integrity: all defined audit rules load and generate events	`custom.rules` contains `-F arcch=b64` (typo for `arch`) and a 32-bit `arch=b32` execve rule. `auditctl -l` showed only the `passwd_changes` rule. `ausearch -k program_execution` and `-k auth_failures` returned `<no matches>`.	System Administrator, with Security Operations review	**Not effective.** The execution-monitoring rule is not working, and the shadow and auth rules are not confirmed loaded.	KPI: 100% of rules in `/etc/audit/rules.d/` appear in `auditctl -l`. KRI: any key with zero events after a known test action.	High. Monitoring is configured on paper but silently failing, which gives false assurance. Process-execution visibility is a core detection control.	Correct `arcch` to `arch`, and remove or justify the b32 rule on a 64-bit-only system. Run `augenrules --load`, restart `auditd`, and check for load errors.	Run `auditctl -l` and confirm every rule is present. Run a test command, then `ausearch -k program_execution` and confirm hits.
2. Audit and log coverage for authentication: failed logins are detectable and retained	`-w /var/log/auth.log` watch returned no events. `grep -i "failed password"` against `/journalctl` failed with "No such file or directory". `aureport --login` shows no events of interest. The system uses journald, not a flat auth log.	System Administrator	**Gap.** Authentication monitoring depends on a file that does not exist, so a zero-failure result cannot be trusted.	KRI: the failed-login count is only valid if the data source is verified. Alert threshold (suggested): 5 failed logins per account within 10 minutes.	Medium to High. Brute-force attempts could go unrecorded while reports show "0 failed logins".	Replace the file watch with `journalctl -u ssh` / `_COMM=sudo` queries, or enable rsyslog to create `auth.log`. Keep auditd's user-auth record types (`USER_AUTH`, `USER_LOGIN`) active.	Generate a deliberate failed `su` or `ssh` attempt, then confirm it appears in `aureport --failed` and the journal.
3. Privileged access to sensitive files: access to `/etc/passwd` is authorised, attributable and reviewed	Audit event 18:22:50, `key="passwd_changes"`: `auid=1000`, `euid=0`, `exe=/usr/bin/sudo`, proctitle decodes to `sudo nano /etc/passwd`. Journal shows `sudo ... COMMAND=/usr/bin/journalctl -f` at 18:35:32. `aureport` shows 0 account changes.	System Administrator (owner of privileged use), with Security Officer for review	**Detected and attributable.** Access is logged, but it is unclear whether it was authorised or whether the file was changed.	KPI: 100% of privileged access to identity files has a documented reason. KRI: any `passwd_changes` event without a matching change ticket.	Medium. Editing account files with root rights is a persistence and privilege-escalation route. Opening it in an editor risks accidental modification.	Use `vipw` or read-only viewing (`less`) instead of `nano`. Record a change ticket for any edit. Compare file hashes against a baseline.	Compare `sha256sum /etc/passwd` to the baseline. Check `aureport -m` for account changes. Review `passwd_changes` events weekly.

Control / Objective	Evidence	Owner	Observed Status	KPI/KRI or Threshold	Risk / Significance	Remediation	Retest / Follow-up
4. Patch and package hygiene: updates are current and repositories are trusted	`apt update` reports **2,638 packages can be upgraded**. The Elastic repository failed signature verification (SHA1 `is not considered secure since 2026-01-01`), so its index was ignored.	System Administrator	**Overdue.** Large patch backlog, and one third-party repo is not being updated.	KPI: critical or security patches applied within 14 days (suggested). KRI: more than 0 repos failing verification, or a backlog above an agreed threshold.	High. An unpatched system carries known vulnerabilities. A repo that cannot be verified cannot be trusted to deliver updates.	Run a full upgrade after snapshotting the VM. Fix or remove the Elastic repo (update its signing key or drop it). Do not bypass the signature check.	Re-run `apt update` with no warnings, and check that `apt list --upgradable` is near zero. Re-run a Lynis scan to compare scores.
5. Log centralisation and unnecessary services: logs are forwarded and only required services run	`auditd` started with "**No plugins found, not dispatching events**". `auditd` was initially absent (`Unit auditd.service could not be found`). Recurring journal errors from `obexd` (Bluetooth OBEX), the `gkr-pam` login module and `vmwgfx` appear on every boot since May 10. The VBox shared clipboard is active.	System Administrator	**Partial.** Logs exist locally, but nothing is forwarded. Unneeded services and recurring unresolved errors remain.	KPI: 100% of security logs forwarded to a central store. KRI: the same error repeated across more than 3 boots without a ticket.	Medium. Local-only logs can be altered by a compromised host. Extra services expand the attack surface, and repeated errors hide real signals.	Configure the audisp/syslog plugin to forward to a central collector. Disable `obex` and the clipboard channel if unused. Fix the VM graphics controller and the PAM keyring config.	Confirm events arrive at the collector. Run `systemctl list-units --state=running` against an approved baseline. Check `journalctl -p err -b` for recurrence.

Governance Escalation Questions

1. Which finding is the highest priority, and why?

The highest priority is Finding 1, audit rule integrity, closely followed by the patch backlog (Finding 4).

- Finding 1 undermines every other control. `custom.rules` has the `arcch=b64` typo, the execve rule produced `<no matches>`, and only `passwd_changes` appeared in `auditctl -l`. Reports showing "0 failed logins" and "0 account changes" look reassuring, but they cannot be trusted while the detection layer is not confirmed to work. This is false assurance, which is worse than a known gap because nobody is looking for it.
- It is also the cheapest and fastest to fix: correct the typo, reload the rules, and test.
- Finding 4 ranks second on exposure. 2,638 pending upgrades and an untrusted Elastic repository mean known vulnerabilities remain open. Rank it first instead if your course frames priority by exploitability rather than by control failure.

2. Who should own remediation?

Finding	Remediation owner	Accountable / reviewer
1. Audit rule integrity	System Administrator	Security Operations (verifies rules fire)
2. Authentication log coverage	System Administrator	Security Operations / SOC lead
3. Privileged access to `/etc/passwd`	System Administrator (the user holding sudo)	Security Officer (reviews access against change tickets)
4. Patch and repository hygiene	System Administrator / IT Operations	Risk function (accepts any exceptions)
5. Log centralisation and unnecessary services	System Administrator, with the logging or platform team	CISO (owns the logging standard)

Separate the person who fixes from the person who verifies. A sysadmin should not be the only one to confirm that their own audit rules work. Replace these roles with names from your course scenario.

3. Escalation triggers

The thresholds below are suggested practice values, not requirements from your materials. Align them with your policy.

Escalate to	Trigger
CISO	Any audit rule or key that is still non-functional 7 days after detection. Any `passwd_changes` or `shadow_changes` event with no matching change ticket. Logs not forwarded centrally on a system in scope for compliance. Any critical Lynis warning left open past its deadline.

Escalate to	Trigger
Risk function	Patch backlog above the agreed threshold, or security patches older than 14 days. A repository failing signature verification for more than one cycle. A control that cannot be fixed and needs a formal risk acceptance with an expiry date.
Management committee	Evidence of unauthorised account changes or a confirmed brute-force pattern (for example, 5 or more failed logins per account within 10 minutes, followed by a success). Repeated missed remediation deadlines across reporting periods. A monitoring failure affecting a regulated or business-critical system.

4. Evidence of successful remediation

- **Finding 1:** `auditctl -l` lists every rule in `/etc/audit/rules.d/`. `augenrules --load` shows no errors. After a test command, `ausearch -k program_execution` returns hits.
- Finding 2: a deliberate failed `su` or `ssh` attempt appears in both `journalctl` and `aureport --failed`. A documented log source replaces the missing `auth.log`.
- Finding 3: a change ticket for each access event, plus a `sha256sum /etc/passwd` match against the baseline and `aureport -m` showing no unexplained account changes.
- **Finding 4:** `apt update` completes without signature warnings, `apt list --upgradable` is near zero, and the Elastic repo is either re-signed or removed. A before and after Lynis hardening index makes this easy to show.
- Finding 5: events arrive at the central collector, the `auditd` "No plugins found" message is gone, and `systemctl list-units --state=running` matches an approved baseline.

Evidence should be dated, taken as screenshots or command output, and reviewed by someone other than the person who fixed the issue.

5. When and how to retest

- Immediately after the fix: run a functional test that generates the event on purpose and confirms it is captured. This is a positive control test, and it matters more than checking that a service is "running".
- Within 7 days: an independent reviewer repeats the test, to confirm the fix survived a reboot and a rule reload.
- **Ongoing:**
    - Weekly: review `passwd_changes` events and zero-event keys.
    - Monthly: compare patch status, run Lynis, and compare the hardening index with the previous run.
    - Quarterly: sample test the whole control set and report results to the Risk function.
- Method: re-run the same commands from the evidence list, record before and after results, and close the finding only when a reviewer signs off. If the retest fails, reopen the finding and apply the escalation triggers above.